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 Source: [github.com/stdgraph/graph-v2](https://github.com/stdgraph/graph-v2)

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# Revision History

## P1709R4

This was a major redesign that incorporated all the experience and input from the past four years.

- Revisit the algorithms to be considered.
- Reduce the scope to focus on an adjacency list with outgoing edges, edge list, and remove mutable interface functions.
- Replace directed and undirected concepts with overridable types of `unordered_edge` for a graph type.
- Simplify the Graph Container types and functions. In particular, `const` and non-`const` variations were consolidated to a single definition to handle both cases when appropriate.
- All Graph Container Interface functions are customization points.
- Introduce Views, inspired by NWGraph design, resulting in simpler and cleaner interfaces to traverse a graph, and simplifying the container interface design.
- Add support for bipartite and multipartite graphs.
- Replace the two container implementations with `compressed_graph`, based on the Compressed Sparse Row matrix, a commonly used data structure for high-performance graphs.

## P1709R3

A simple status revision to say a major change is coming soon.

## P1709R2

Define the **uniform API** for undirected and directed algorithms (an extended API also exists for directed graphs). Added **concepts** for undirected, directed and bidirected graphs. Refined **DFS** and **BFS** range definitions from prototype experience. Refined **shortest paths** and **transitive closure** algorithms from input and prototype experience.

## P1709R1

Rewrite with a focus on a **purely functional design**, emphasizing the algorithms and graph API. Also added **concepts** and **ranges** into the design. Addressed concerns from Cologne review to change to functional design.

## P1709R0

Focus on **object-oriented API** for data structures and example code for a few algorithms.

# Chapter1 Overview

Graphs, used in ML and other **scientific** domains, as well as **industrial** and **general** programming, do **not** presently exist in the C++ standard. In ML, a graph forms the underlying structure of an **artificial neural network** (ANN). In a **game**, a graph can be used to represent the **map** of a game world. In **business** environments, graphs arise as **entity relationship diagrams** (ERD) or **data flow diagrams** (DFD). In the realm of **social media**, a graph represents a **social network**.

This document proposes the addition of **graph algorithms**, **graph views**, **graph container interface** and a **graph container implementation** to the C++ library to support **machine learning** (ML), as well as other applications. ML is a large and growing field, both in the **research community** and **industry**, that has received a great deal of attention in recent years. This paper presents an **interface** of the proposed algorithms, views, graph functions and containers.

## 1.1 Goals and Priorities

- Follow the separation of algorithms, ranges, views and containers established by the standard library.
- Graph algorithms have the following characteristics
  - Support syntax that is simple, expressive and easy to understand. This should not compromise the ability to write high-performance algorithms.
  - Vertices are required to be in random access containers with an integral `vertex_id` in this proposal.
- Graph views provide common traversals of a graph's vertices and edges that is more concise and consistent than using the graph container interface directly. They include simple traversals like `vertexlist` (all vertices in the graph) and `incidence edges` (edges on a vertex), as well as more complex traversals like depth-first and breath-first searches.
- All free functions are customization point objects, unless noted otherwise. Reasonable default implementations are provided whenever possible.
- The Graph Container Interface provides a consistent interface that can be used by algorithms and views. It has the following characteristics:
  - The interface models an adjacency graph container, which is an outer range of vertices with an inner range of outgoing (a.k.a. incidence) edges on each vertex.
  - Definition of concepts, types, type traits, type aliases, and functions used by algorithms and views.
    - Type traits will be defined that can be overridden for each graph container to give additional hints that can be used by algorithms to refine their behavior, such as `adjacency_matrix` and `unordered_edge`.
  - Support of optional user-defined value types on an edge, vertex and/or the graph itself.
  - Support bipartite and multipartite graphs, as long as the underlying graph supports it. If the underlying graph doesn't support either, it is considered unipartite with a single partition.
  - Allow for useful extensions of the graph data model in future proposals or in external graph implementations.
- Define an Edge List interface, required by some algorithms, that can be used by user-defined ranges for algorithms that require them.
- Provide an initial suite of useful functionality that includes algorithms, views, container interface, and at least one model container implementation.

## 1.2 Examples

The following code demonstrates how a simple graph can be created as a range of ranges, using the standard containers.

[PHIL: Need "basic" bfs so we don't have to include `uv`]

[PHIL: Duplicated in Introduction]

```
std::vector<std::string> actors { "Tom Cruise", "Kevin Bacon", "Hugo Weaving",
                                "Carrie-Anne Moss", "Natalie Portman", "Jack Nicholson",
                                "Kelly McGillis", "Harrison Ford", "Sebastian Stan",
                                "Mila Kunis", "Michelle Pfeiffer", "Keanu Reeves",
```

```

        "Julia Roberts" });

using G = std::vector<std::vector<int>>>;
auto target_id(const G& g, edge_reference_t<G> uv) {return get<0>(uv);}
G costar_adjacency_list{
    {1, 5, 6}, {7, 10, 0, 5, 12}, {4, 3, 11}, {2, 11}, {8, 9, 2, 12}, {0, 1}, {7, 0},
    {6, 1, 10}, {4, 9}, {4, 8}, {7, 1}, {2, 3}, {1, 4} };

int main() {
    std::vector<int> bacon_number(size(actors));

    // 1 -> Kevin Bacon
    for (auto&& [uid,vid,uv] : sourced_edges_breadth_first_search(costar_adjacency_list, 1)) {
        bacon_number[vid] = bacon_number[uid] + 1;
    }

    for (int i = 0; i < size(actors); ++i) {
        std::cout << actors[i] << " has Bacon number " << bacon_number[i] << std::endl;
    }
}

```

`target_id(g, uv)` defines the required function to get a `target_id` for an edge in the graph `G`. Other functions can also be overridden to allow a developer to adapt their own graph data structures to the library.

### 1.3 What this proposal is not

This paper limits itself to adjacency graphs and edgelist only. An adjacency graph is an outer range of vertices with an inner range of outgoing edges on each vertex. An edgelist is a view of edges, which is either all the edges in the adjacency graph or a projection of a user-defined range.

Parallel versions of the algorithms are not included for several reasons. The executors proposal in P2300r5 [1] is expected to introduce new and better ways to do parallel algorithms beyond that used in the parallel STL algorithms and we would like to wait for finalization of that proposal before committing to parallel implementations. Secondly, many graph algorithms don't benefit from parallel implementations so there is less need to offer an implementation. Lastly, it will help limit the size of this proposal which is already looking to be large without it. It is expected that future proposals will be submitted for parallel graph algorithms.

Incoming edges on a vertex are not included, though it is hoped that a future proposal will be made for them.

The algorithms and views in this proposal expect that `vertex_ids` are densely assigned in a random access range, but it does not exclude the possibility of sparsely-defined `vertex_ids` stored in containers like `std::map` or `std::unordered_map` in future proposals.

The algorithms and views in this proposal expect that `vertex_ids` are integral, but it does not exclude non-integral or user-defined types in future proposals.

Hypergraphs are not supported.

### 1.4 Impact on the Standard

This proposal is a pure **library** extension.

### 1.5 Interaction with Other Papers

There is no interaction with other proposals to the standard.

### 1.6 Implementation Experience

The github [github.com/stdgraph](https://github.com/stdgraph) repository contains an implementation for this proposal.

### 1.7 Usage Experience

There is no current use of the library. There are plans to begin using it in the next year in a commercial setting.

## 1.8 Deployment Experience

There is no current deployment experience of the library. There are plans for this to follow the usage experience.

## 1.9 Performance Considerations

The algorithms are being ported from NWGraph to the [github.com/stdgraph](https://github.com/stdgraph) implementation used for this proposal. Performance analysis from those algorithms can be found in the peer-reviewed papers for NWGraph [2, 3].

### 1.10 Prior Art

**boost::graph** has been an important C++ graph implementation since 2001. It was developed with the goal of providing a modern (at the time) generic library that addressed all the needs someone would want of a graph library. It is still a viable library used today, attesting to the value it brings.

However, boost::graph was written using C++98 in an “expert-friendly” style, adding many abstractions and using sophisticated template metaprogramming, making it difficult to use by a casual developer.

(Andrew is a co-author of boost::graph.)

**NWGraph** ([4] and [2]) was published in 2022 by Lumsdaine et al, bringing additional experience gained since creating boost::graph, to create a modern graph library using C++20 for its implementation that was more accessible to the average developer.

While NWGraph made important strides to introduce the idea of the graph as a range-of-ranges and implemented many important algorithms, there are some areas it didn’t address that come a practical use in the field. For instance, it didn’t have a well-defined API for graph data structures that could be applied to existing graphs, and there wasn’t a uniform approach to properties.

This proposal takes the best of NWGraph, with previous work done for P1709 to define a Graph Container Interface, to provide a library that embraces performance, ease-of-use and the ability to use the algorithms and views on externally defined graph containers.

### 1.11 Alternatives

There are no known alternative graph library we’re aware of that meets the same requirements and uses concepts and ranges from C++20.

### 1.12 Feature Test Macro

The `__cpp_lib_graph` feature test macro is recommended to represent all features in this proposal including algorithms, views, concepts, traits, types, functions and graph container(s).

### 1.13 Freestanding

We believe this library can be used in a freestanding C++ implementation.

### 1.14 Namespaces

Graph containers and their views and algorithms are not interchangeable with existing containers and algorithms. Additionally, there are some domain-specific terms that may clash with existing or future names, such as `degree` and `partition_id`. For these reasons, we recommend their own namespaces as follows. This assumption is used in this proposal.

```
std::graph
std::graph::views
```

Alternative locations for the above respective namespaces could also be as follows:

```
std::ranges
std::ranges::views
```

# Chapter2 Introduction

## 2.1 Motivation

The original STL revolutionized the way that C++ programmers could apply algorithms to different kinds of containers, by defining *generic* algorithms, realized via function templates. A hierarchy of *iterators* were the mechanism by which algorithms could be made generic with respect to different kinds of containers, Named requirements specified the valid expressions and associated types that algorithms required of their arguments. As of C++20, we now have both ranges and concepts, which now provide language-based mechanisms for specifying requirements for generic algorithms.

As powerful as the algorithms in the standard library are, the underlying basis for them is a range (or iterator pair), which inherently can only specify a one-dimensional container. Iterator pairs (equiv. ranges) specify a `begin()` and an `end()` and can move between those two limits in various ways, depending on the type of iterator. As a result, important classes of problems that programmers are regularly faced with use structures that are not one-dimensional containers, and so the standard library algorithms can't be directly used. Multi-dimensional arrays are an example of one such kind of data structure. Matrices do have the nice property that they (typically) have the ability to be “raveled”, i.e., the data underlying the matrix can still be treated as a one-dimensional container. Multi-dimensional arrays also have the property that, even though they can be thought of as hierarchical containers, the hierarchy is uniform—an N-dimensional array is a container of N-1 dimensional arrays.

Another important problem domain that does not fit into the category of one-dimensional ranges is that of *graph algorithms and data structures*. Graphs are a powerful abstraction for modeling relationships between entities in a given problem domain, irrespective of what the actual entities are, and irrespective of what the actual relationships are. In that sense, graphs are, by their very nature, generic. Graphs are a fundamental abstraction in computer science, and are ubiquitous in real-world applications.

Any problem concerned with connectivity can be modeled as a graph. Just a small set of examples include Internet routing, circuit partitioning and layout, finding the best route to take to a destination on map. There are also relationships between entities that are inferred from large sets of data, for example the graph of consumers who have purchased the same product, or who have viewed the same movie. Yet more interesting structures arise (hypergraphs or k-partite graphs) can arise when we want to model relationships between diverse types of data, such as the graph of consumers, the products they have purchased, and the vendors of the products. And, of course, graphs play a critical role in multiple aspects of machine learning.

On the flip side of graph structures are the graph algorithms that are widely used for problems such as the above. Well-known graph algorithms include breadth-first search, Dijkstra's algorithm, connected components, and so on. Because graphs can come from so many different problem domains, they will also be represented with many different kinds of data structures. To make graph algorithms as usable as possible across arbitrary representation requires application of the same principles that were used in the original STL: a collection of related algorithms from a problem domain (in our case, graphs), minimizing the requirements imposed by the algorithms on their arguments, systematically organizing the requirements, and realizing this framework of requirements in the form of concepts.

There are also many uses of graphs that would not be met by a standard set of algorithms. A standardized interface for graphs is eminently useful in such situations as well. In the most basic case, it would provide a well-defined framework for development. But in keeping with the foundational goal of generic programming to enable reuse, it would also empower users to develop and deploy their own reusable graph components. In the best case, such algorithms would be available to the broader C++ programmer community.

Because graphs are so ubiquitous and so important to modern software systems, a standardized library of graph algorithms and data structures would have enormous benefit to the C++ development community. This proposal contains the specification of such a library, developed using the principles above.

## 2.2 Example: Six Degrees of Kevin Bacon

A classic example of the use of a graph algorithm is the game “The Six Degrees of Kevin Bacon.” The game is played by connecting actors to each other through movies they have appeared in together. The goal is to find the smallest number of movies that connect a given actor to Kevin Bacon. That number is called the “Bacon number” of the actor. Kevin Bacon himself has a Bacon number of 0. Since Kevin Bacon appeared with Tom Cruise in “A Few Good Men”, Tom Cruise has a Bacon number of 1.

The following program computes the Bacon number for a small selection of actors.

```
std::vector<std::string> actors { "Tom Cruise", "Kevin Bacon", "Hugo Weaving",
    "Carrie-Anne Moss", "Natalie Portman", "Jack Nicholson",
    "Kelly McGillis", "Harrison Ford", "Sebastian Stan",
    "Mila Kunis", "Michelle Pfeiffer", "Keanu Reeves",
    "Julia Roberts" };

using G = std::vector<std::vector<int>>>;
auto target_id(const G& g, edge_reference_t<G> uv) {return get<0>(uv);}
G costar_adjacency_list{
    {1, 5, 6}, {7, 10, 0, 5, 12}, {4, 3, 11}, {2, 11}, {8, 9, 2, 12}, {0, 1}, {7, 0},
    {6, 1, 10}, {4, 9}, {4, 8}, {7, 1}, {2, 3}, {1, 4} };

int main() {
    std::vector<int> bacon_number(size(actors));

    // 1 -> Kevin Bacon
    for (auto&& [uid,vid,uv] : sourced_edges_breadth_first_search(costar_adjacency_list, 1)) {
        bacon_number[vid] = bacon_number[uid] + 1;
    }

    for (int i = 0; i < size(actors); ++i) {
        std::cout << actors[i] << " has Bacon number " << bacon_number[i] << std::endl;
    }
}
```

Output:

```
Tom Cruise has Bacon number 1
Kevin Bacon has Bacon number 0
Hugo Weaving has Bacon number 3
Carrie-Anne Moss has Bacon number 4
Natalie Portman has Bacon number 2
Jack Nicholson has Bacon number 1
Kelly McGillis has Bacon number 2
Harrison Ford has Bacon number 1
Sebastian Stan has Bacon number 3
Mila Kunis has Bacon number 3
Michelle Pfeiffer has Bacon number 1
Keanu Reeves has Bacon number 4
Julia Roberts has Bacon number 1
```

In graph parlance, we are creating a graph where the vertices are actors and the edges are movies. The number of movies that connect an actor to Kevin Bacon is the shortest path in the graph from Kevin Bacon to that actor. In the example above, we compute shortest paths from Kevin Bacon to all other actors and print the results. Note, however, that actor-actor relationships are not how data about actors is available in the wild (from IMDB, for example). Rather, two types of relationships available are actor-movie and movie-actor. See Section ?? below.

## 2.3 Graph Background

For clarity, we briefly review some of the basic terminology of graphs. We use commonly accepted terminology for graph data structures and algorithms and adopt the particular terminology used in the textbook by Cormen, Leiserson, Rivest, and Stein (“CLRS”) [5].

### 2.3.1 Basic Terminology

To model the relationships between entities, a *graph*  $G$  comprises two sets: a *vertex set*  $V$ , whose elements correspond to the entities, and an *edge set*  $E$ , whose elements are pairs corresponding to elements in  $V$  that have some relationship with each other. That is, if  $u$  and  $v$  are members of  $V$  that have some relationship that we wish to capture, then there is a pair  $\{u, v\}$  in  $E$ . We can express that together  $V$  and  $E$  define a graph as  $G = \{V, E\}$ .



Two examples of graph models are shown in Figures ?? and ??, which respectively model a network of routes between and an electronic circuit. The figures show the domain-specific data to be modeled and the sets  $V$  and  $E$  for each graph. Also shown for each graph is a node and link diagram, a commonly-used graphical<sup>1</sup> notation.

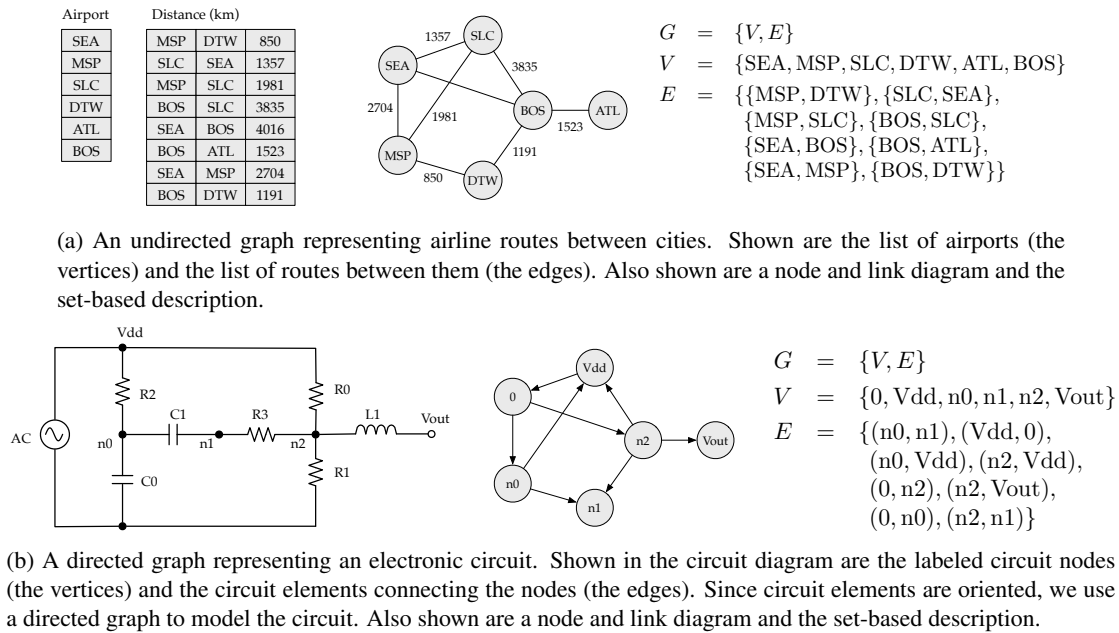


Figure 2.1 — Graph models of an airline route system and of an electronic circuit.

### 2.3.2 Graph Representation: Enumerating the Vertices

To reason about graphs, and to write algorithms for them, we require a *representation* of the graph. We note that *a graph and its representation are not the same thing*. It is therefore essential that we be precise about this distinction as we develop a software library of graph algorithms and data structures<sup>2</sup>.

The representations that we will be using are familiar ones: adjacency matrix, edge list, and adjacency list. We begin with a process that is so standard that we typically don't even notice it, but it forms the foundation of graph representations: we *enumerate the vertices*. That is, we assign an index to each element of  $V$  and write  $V = \{v_0, v_1, \dots, v_{n-1}\}$ . Based on that enumeration, elements of  $E$  are expressed in the form  $\{v_i, v_j\}$ . Similarly, we can enumerate the edges, and write  $E = \{e_0, e_1, \dots, e_{m-1}\}$ , though the enumeration of  $E$  does not play a role in standard representations of graphs. The number of elements in  $V$  is denoted by  $|V|$  and the number of elements in  $E$  is denoted by  $|E|$ .

We summarize some remaining terminology about vertices and edges.

- An edge  $e_k$  may be *directed*, denoted as the ordered pair  $e_k = (v_i, v_j)$ , or it may be *undirected*, denoted as the (unordered) set  $e_k = \{v_i, v_j\}$ . The edges in  $E$  are either all directed or all undirected, corresponding respectively to a *directed graph* or to an *undirected graph*.
- If the edge set  $E$  of a directed graph contains an edge  $e_k = (v_i, v_j)$ , then vertex  $v_j$  is said to be *adjacent* to vertex  $v_i$ . The edge  $e_k$  is an *out-edge* of vertex  $v_i$  and an *in-edge* of vertex  $v_j$ . Vertex  $v_i$  is the *source* of edge  $e_k$ , while  $v_j$  is the *target* of edge  $e_k$ .
- If the edge set  $E$  of an undirected graph contains an edge  $e_k = \{v_i, v_j\}$ , then  $e_k$  is said to be *incident* on the vertices  $v_i$  and  $v_j$ . Moreover, vertex  $v_j$  is adjacent to vertex  $v_i$  and vertex  $v_i$  is adjacent to vertex  $v_j$ . The edge  $e_k$  is an out-edge of both  $v_i$  and  $v_j$  and it is an in-edge of both  $v_i$  and  $v_j$ .
- The *neighbors* of a vertex  $v_i$  are all the vertices  $v_j$  that are adjacent to  $v_i$ . The set of all of the neighbors is the *neighborhood* of  $v_i$ .

<sup>1</sup> An unfortunate collision of terminology.

<sup>2</sup> In fact, if we are to be completely precise, the library we are proposing is one of algorithms and data structures for graph representations. We will make concessions to commonly accepted terminology, while precisely defining that terminology.

- A *path* as a sequence of vertices  $v_0, v_1, \dots, v_{k-1}$  such that there is an edge from  $v_0$  to  $v_1$ , an edge from  $v_1$  to  $v_2$ , and so on. That is, a path is a set of edges  $(v_i, v_{i+1}) \in E$  for  $i = 0, 1, \dots, k - 2$ .

### 2.3.3 Adjacency-Based Representations

We begin our development of graph representations with the almost universally-accepted definition of the adjacency matrix representation of a graph. The *adjacency matrix representation* of a graph  $G$  is a  $|V| \times |V|$  matrix  $A = (a_{ij})$  such that, respectively for a directed or undirected graph

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases} \quad a_{ij} = a_{ji} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases}$$

That is,  $a_{ij} = 1$  if and only if  $v_j$  is adjacent to  $v_i$  in the original graph  $G$  (hence the name “adjacency matrix”). Here we can see why we said that the initial enumeration of  $V$  is foundational to representations: *The adjacency matrix is based solely on the indices used in that enumeration.* It does not contain the vertices or edges themselves.

As a data structure to use for algorithms, the adjacency matrix is not very efficient, neither in terms of storage (which, at  $|V| \times |V|$  is prohibitive), nor for computation. Instead of storing the entire adjacency matrix, we can simply store the index values of its non-zero elements. A *sparse coordinate adjacency matrix* is a container  $C$  of pairs  $(i, j)$  for every  $a_{ij}$  in  $A$ . At first glance, it may seem that we have simply created a data structure  $C$  that has a pair  $(i, j)$  if  $E$  in the original graph has an edge from  $v_i$  to  $v_j$ . This is true in the directed case. However, in the undirected case, if there is an edge between  $v_i$  and  $v_j$ , then  $v_i$  is adjacent to  $v_j$  and  $v_j$  is adjacent to  $v_i$ . In other words, if there is an edge between  $v_i$  and  $v_j$  in an undirected graph, then both the entries  $a_{ij}$  and  $a_{ji}$  are equal to 1<sup>3</sup> — and therefore for a single edge between  $v_i$  and  $v_j$ ,  $C$  contains two index pairs:  $(i, j)$  and  $(j, i)$ . The sparse coordinate representation is commonly known as *edge list*. However, we caution the reader that  $C$  does not store edges, but rather indices and that, in the case that it represents an undirected graph, there is not a 1-1 correspondence between the edges in  $E$  and the contents of  $C$ .

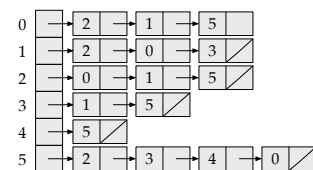
Although the sparse coordinate adjacency matrix is much more efficient in terms of storage than the original adjacency matrix, it isn’t as efficient as it could be. Much more importantly, it does not at all useful for the types of operations used by most graph algorithms, which need to be able to get the set of neighbors of a given vertex in constant time. To support this type of operation, we use a *compressed sparse adjacency matrix*, which is an array  $J$  with  $|V|$  entries, where each  $J[i]$  is a linear container of indices  $\{j\}$  such that  $v_j$  is a neighbor of  $v_i$  in  $G$ . That is  $j$  is contained in  $J[i]$  if and only if there is an edge  $(v_i, v_j)$  in  $E$  (or, equivalently, if there is a pair  $(i, j)$  in  $C$  or, equivalently, if  $a_{ij} = 1$ )<sup>4</sup>. We note that if  $(v_i, v_j)$  is an edge in an undirected graph,  $J[i]$  will contain  $j$  and  $J[j]$  will contain  $i$ . The common name for this data structure is *adjacency list*. Although this name is problematic (for instance, it is not actually a list), it is so widely used that we also use it here—but *we mean specifically that an “adjacency list” is the compressed sparse adjacency matrix representation of a graph*<sup>5</sup>. Again we emphasize the distinction between a graph and its representation: An adjacency list  $J$  is not the same as the graph  $G$ —it is a representation of  $G$ .

Illustrations of the adjacency-matrix representations of the airline route graph and the electronic circuit graph are shown in Figures ?? and ??, respectively.

|     |   |
|-----|---|
| ATL | 4 |
| BOS | 5 |
| DTW | 3 |
| MSP | 1 |
| SEA | 0 |
| SLC | 2 |

$$A = \begin{bmatrix} & & & & & \\ & 1 & 1 & & & 1 \\ & & & 1 & 1 & \\ 1 & & & & & 1 \\ & 1 & & & & 1 \\ & & 1 & & & \\ & & & & & 1 \\ 1 & & & 1 & 1 & 1 \end{bmatrix}$$

|   |   |
|---|---|
| 3 | 1 |
| 1 | 3 |
| 2 | 1 |
| 1 | 2 |
| 5 | 0 |
| 0 | 5 |
| 1 | 0 |
| 0 | 1 |
| 2 | 0 |
| 0 | 2 |
| 5 | 2 |
| 2 | 5 |
| 5 | 4 |
| 4 | 5 |
| 5 | 3 |
| 3 | 5 |



(a) An enumeration of the airport graph given in 2.1a.

(b) The adjacency matrix representation of the graph given in Figure 2.1a, using the enumeration given in Figure 2.2a.

(c) The coordinate sparse adjacency matrix representation (shown split into two columns).

(d) The compressed sparse adjacency matrix representation.

Figure 2.2 — Adjacency matrix representations of the airport graph model.

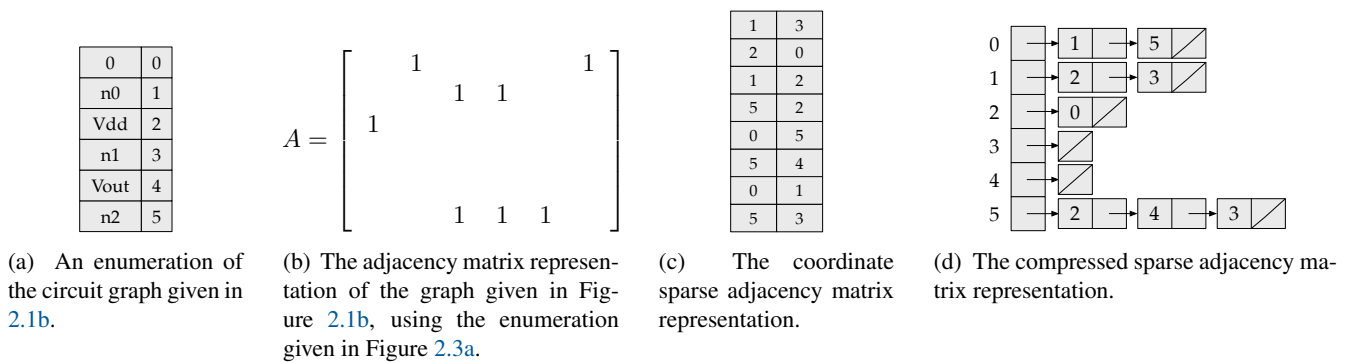


Figure 2.3 — Adjacency matrix representations of the circuit graph model.

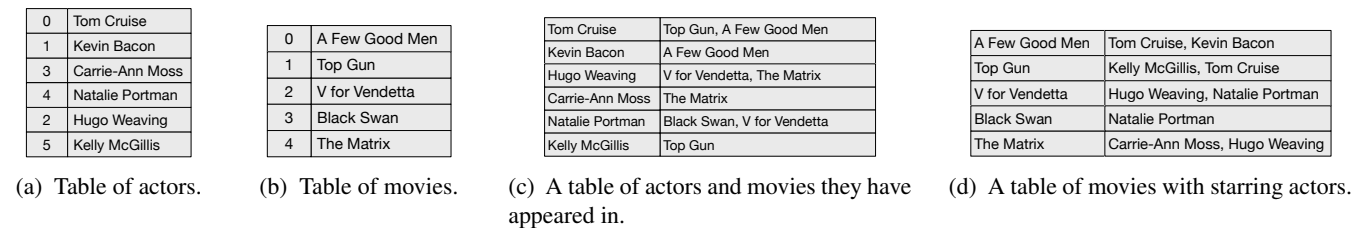


Figure 2.4 — Illustrative simplification of IMDB actor and movie data.

## 2.4 Bipartite Graphs

So far, we have been considering graphs where edges in  $E$  are pairs of vertices, which are taken from a single set  $V$ . We refer to such a graph as a *unipartite* graph. But consider again the Kevin Bacon example. The source for the information comprising the Kevin Bacon data is the Internet Movie Database (IMDB). However, the IMDB does not contain any explicit information about the relationships between actors. Rather it contains files of tabular data, one of which contains an entry for each movie with the list of actors that have appeared in that movie, and another of which contains an entry for each actor with the list of movies that actor has appeared in (“movie-actor” and “actor-movie” tables, respectively). Such tables are shown in Figure 2.4.<sup>6</sup> Thus, a graph, as we have defined it, cannot model the IMDB.

There is a small generalization we can make to the definition of graph that will result in a suitable abstraction for modeling the IMDB. In particular, we need one set of vertices corresponding to actors, another set of vertices corresponding to movies, and then a set of edges corresponding to the relationships between actors and movies. There are two kinds of relationships to consider actors in movies or movies starring actors. To be well-defined, the edge set may only contain one kind of relationship. To capture this kind of model, we define a *structurally bipartite graph*  $H = \{U, V, E\}$ , where vertex sets  $U$  and  $V$  are enumerated  $U = \{u_0, u_1, \dots, u_{n_0}\}$  and  $V = \{v_0, v_1, \dots, v_{n_1}\}$ , and the edge set  $E$  consists of pairs  $(u_i, v_j)$  where  $u_i$  is in  $U$  and  $v_j$  is in  $V$ .

The *adjacency matrix representation* of a structurally bipartite graph is a  $|U| \times |V|$  matrix  $A = (a_{ij})$  such that,

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases}$$

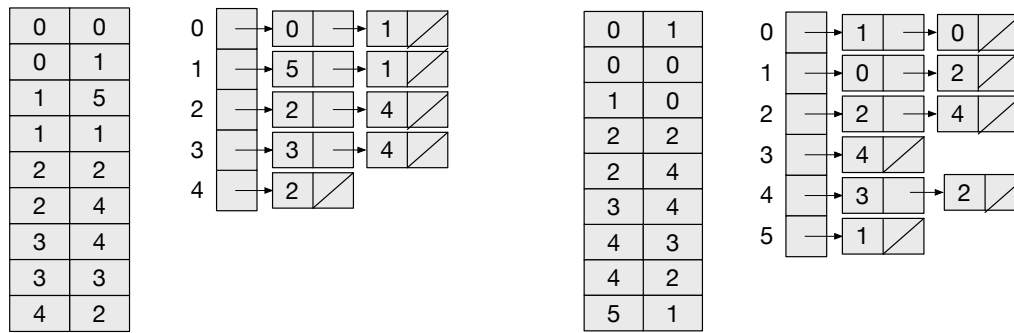
From this adjacency matrix representation we can readily construct coordinate and compressed sparse representations. The only structural difference between the representations of a structurally bipartite graph and that of a unipartite graph is that of vertex cardinality. That is, in a unipartite graph, edges map from  $V$  to  $V$ , and hence the values in the left hand column and in the right hand column of a coordinate representation would be in the same range:  $[0, |V|)$ . However, for a structurally bipartite graph, this is no longer the case. Although the coordinate representation still consists of pairs of vertex indices, the range of values in the left hand column is  $[0, |U|)$ , while in the right hand column it is  $[0, |V|)$ . Similarly, the compressed representation will have  $|U|$  entries, but the values stored in each entry may range from  $[0, |V|)$ . We note that these are constraints on values, not on structure.

<sup>3</sup>That is, the adjacency matrix is symmetric.

<sup>4</sup>The compressed sparse adjacency matrix is identical to the compressed sparse row format from linear algebra

<sup>5</sup>We concede that “adjacency list” rolls off the tongue much more easily than “compressed sparse adjacency matrix representation of a graph.”

<sup>6</sup>This is a greatly simplified version of the CSV files that actually comprise the IMDB. The full set of files is available for non-commercial use at <https://datasets.imdbws.com>.



(a) Coordinate and compressed sparse adjacency representations for movies with their starring actors.

(b) Coordinate and compressed sparse adjacency representations for actors and the movies they have appeared in.

Figure 2.5 — Sparse adjacency representations (edge lists and adjacency lists) for IMDB actor and movie data.

We distinguish a structurally bipartite graph from simply a bipartite graph because the former applies separate enumerations to  $U$  and  $V$ . In customary graph terminology, a *bipartite* graph is one in which the vertices can be partitioned into two disjoint sets, such that all of the edges in the graph only connect vertices from one set to vertices of the other set. However, although the vertices are partitioned, they are still taken from the same original vertex set  $V$  and have a single enumeration. Whether a graph can be partitioned in this way is a run-time property inherent to the graph itself (which can be discovered with an appropriate algorithm). This is not a natural way to model separate categories of entities, such as movies and actors, where entities are categorized completely independently of each other and it is therefore most appropriate to have independent enumerations for them. A structurally bipartite graph explicitly captures distinct vertex categories.

## 2.5 Partitioned Graphs

In contrast to structurally bipartite graphs, there are certainly cases where one would want to maintain two categories of entities, or otherwise distinguish the vertices, from the same vertex set. In that case, we would use a *partitioned graph*, which we define as  $G = \{V, E\}$ , where the vertex set  $V$  consists of non-overlapping subsets, i.e.,  $V = \{V_0, V_1, \dots\}$  which we enumerate as  $V_0 = \{v_0, v_1, \dots, v_{n_0-1}\}$ ,  $V_1 = \{v_{n_0}, \dots, v_{n_1-1}\}$  and so on. Each  $V_i$  is a *partition* of  $V$ . The total enumeration of  $V$  is  $V = \{v_0, v_1, \dots, v_{n-1}\}$ . Just as each  $V_i$  is a partition of  $V$ , the enumeration of each  $V_i$  is a partitioning of the enumeration of  $V$ .

The edge set  $E$  still consists of edges  $(v_i, v_j)$  (or  $\{v_i, v_j\}$  where, in general,  $v_i$  and  $v_j$  may come from any partition.

We note that partitioned graphs are not restricted to two partitions—a partitioned graph can represent an arbitrary number of partitions, i.e., a *multipartite* graph (a graph with multiple subsets of vertices such that edges only go between subsets). While partitioned graphs can be used to model multipartite graphs, partitioned graphs are not necessarily multipartite; edges can comprise vertices within a partition as well as across partitions.

## 2.6 From Data to Graph

### 2.6.1 Columnar Data

Here we show how one might create an unlabeled edge list from a table of data stored in a CSV file. The following loads a list of directed edges from a CSV file (the values in each row are assumed to be separated by whitespace)<sup>7</sup>. The elements of the first column are considered to be the source vertices and the elements of the second column are the destination vertices. If the edges also had properties, the third column would contain the property values. In this example, the edges are loaded into a vector of tuples, which meets the requirements of a (presumed) `sparse_coordinate` concept.

```
auto sparse_coordinate edges = std::vector<std::tuple<vertex_id_t, vertex_id_t>;
auto input = std::ifstream ("input.csv");
vertex_id_t src, dst;
while (input >> src >> dst) {
    edges.emplace_back (src, dst);
}
```

<sup>7</sup>We take a broad view of what a comma is.

Similarly, we could load a list of undirected edges from a CSV file into a `sparse_coordinate` structure. Note that, as discussed above, the coordinate sparse adjacency matrix representation (aka an edge list), contains an entry  $(i, j)$  as well as an entry  $(j, i)$  for each undirected edge  $\{v_i, v_j\}$ . Hence, we add both  $(src, dst)$  and  $(dst, src)$  to `edges`.

```
auto sparse_coordinate edges = std::vector<std::tuple<vertex_id_t, vertex_id_t, double>
    edges;
auto input = std::ifstream ("input.csv");
vertex_id_t src, dst;
double val;
while (input >> src >> dst >> val) {
    edges.emplace_back (src, dst, val);
    edges.emplace_back (dst, src, val);
}
```

These examples are meant to be illustrative and not necessarily comprehensive (nor efficient). There are, of course, many ways to define containers that meet the requirements of the edge list concept and many ways to create an edge list from columnar data.

## 2.6.2 Converting an Edge List to an Adjacency List

The following creates a compressed sparse representation (an adjacency list) from a coordinate sparse representation. The adjacency list is represented as a `std::vector<std::vector<vertex_id_t>>`;

```
auto sparse_coordinate edges = std::vector<std::tuple<vertex_id_t, vertex_id_t>
// Read the edges
auto sparse_compressed adj_list = std::vector<std::vector<vertex_id_t>>;
for (auto [src, dst] : edges) {
    if (src >= adj_list.size()) {
        adj_list.resize(src + 1);
    }
    adj_list[src].push_back (dst);
}
```

We note that the `sparse_coordinate` representation is agnostic as to whether it was originally created based on directed edges or undirected edges. An optimization to the sparse coordinate representation would be to use a *packed coordinate* representation, which would only maintain a single entry for each undirected edge. In that case, we would need to have two complementary insertions into the adjacency list for each entry in the packed coordinate representation.

The following example illustrates the use of a packed coordinate format to construct an adjacency list with edge property.

```
auto packed_sparse_coordinate edges = std::vector<std::tuple<vertex_id_t, vertex_id_t,
    double>;
// Read the edges
auto compressed_sparse adj_list = std::vector<std::vector<std::tuple<vertex_id_t, double
    >>>(edges.num_vertices());
for (auto [src, dst, val] : edges) {
    adj_list[src].push_back (dst, val);
    adj_list[dst].push_back (src, val);
}
```

## Chapter3 Algorithms

Our proposed set of algorithms are grouped into Tier 1, Tier 2, and Tier 3 (similar to the approach used to prioritize algorithms in `std::ranges`).

Tier 1:

- Breadth-First search
- Dijkstra’s algorithm
- Bellman-Ford
- Triangle counting
- Label propagation
- Articulation points
- Connected components
- Biconnected components
- Strongly connected components
- Topological sort
- Maximal independent set
- Jaccard coefficient
- Kruskal Minimal Spanning Tree
- Prim Minimal Spanning Tree

### 3.1 Introduction

Basic characteristics of the algorithms shown below are summarized in tables of the following form:

|                                               |                                            |                                           |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|

The parts of the table have the following meaning:

- **Complexity** The complexity of the algorithm based on the number of vertices (V) and edges (E).
- **Throws?** Will the algorithm throw at all? If so, look at the *Throws* section after the function prototypes for details.
- **Multi-edge?** Does the algorithm act as expected if more than one edge with the same direction exists between the same two vertices?
- **Cycles?** Does the algorithm act as expected if a vertex (or edge) is part of a cycle?
- **Directed?** Is the algorithm only for directed graphs, or can it also be used for undirected graphs that have complimentary edges, with different directions, between two vertices.

[PHIL: The *Directed?* section needs work.]

### 3.2 Algorithm Concepts

The abstraction that is used for describing and analyzing almost all graph algorithms is the adjacency list. Naturally then implementations of graph algorithms in C++ will operate on a data structure representing an adjacency list. And generic algorithms will be written in terms of concepts that capture the essential operations that a concrete data structure must provide in order to be used as an abstraction of an adjacency list.

Most fundamentally (as illustrated above), an adjacency list is a collection of vertices, each of which has a collection of outgoing edges. In terms of existing C++ concepts, we can consider an adjacency list to be a range of ranges (or, more specifically, a

random access range of forward ranges). The outer range is the collection of vertices, and the inner ranges are the collections of outgoing edges.

```
template <class G, class WF, class DistVal, class Compare, class Combine>
concept basic_edge_weight_function = // e.g. weight(uv)
    is_arithmetic_v<DistVal> &&
    strict_weak_order<Compare, DistVal, DistVal> &&
    assignable_from<add_lvalue_reference_t<DistVal>,
        invoke_result_t<Combine, invoke_result_t<WF, edge_reference_t<G>>>>>;

template <class G, class WF, class DistVal>
concept edge_weight_function = // e.g. weight(uv)
    is_arithmetic_v<invoke_result_t<WF, edge_reference_t<G>>>> &&
    basic_edge_weight_function<G,
        WF,
        Distance,
        less<DistVal>,
        plus<DistVal>>;
```

### 3.3 Shortest Paths

[ANDREW: Note that NetworkX also specifies single source single target and multiple source versions of the shortest paths algorithms. BGL does not have these (nor NWGraph). We should discuss whether or not to consider those and whether or not to make them Tier 1, 2, 3, or infinity.]

#### 3.3.1 Unweighted Shortest Paths: Breadth-First Search

##### 3.3.1.1 Breadth-First Search, Single Source, Initialization

```
template <property P, property D>
void init_breadth_first_search(P&& predecessors, D&& distances);

template <property P, property D>
void init_breadth_first_search(P&& predecessors,
    D&& distances,
    std::ranges::range_value_t<D> init);

template <property D>
void init_breadth_first_distances(D&& distances);

template <property D>
void init_breadth_first_distances(D&& distances, std::ranges::range_value_t<D> init);
```

*Effects:*

- Each `predecessors[i]` is initialized to `i`.
- Each `distance[i]` is initialized to `std::numeric_limits::max()` or to `init`.

##### 3.3.1.2 Breadth-First Search, Single Source

[PHIL: What value does this bring over the `breadth_first_search` view? ]

```
template <adjacency_list_graph G, property D, property P>
void breadth_first_search(const G& graph,
    vertex_id_t<G> source,
    P&& predecessors,
    D&& distances);

template <adjacency_list_graph G, property D, property P, queueable Q>
void breadth_first_search(
    const G& graph, vertex_id_t<G> source, P&& predecessors, D&& distances, Q&& q);
```



```
template <adjacency_list_graph G, property D>
void breadth_first_distances(const G& graph, vertex_id_t<G> source, D&& distances);

template <adjacency_list_graph G, property D, queueable Q>
void breadth_first_distances(const G& graph, vertex_id_t<G> source, D&& distances, Q&& q);
```

1 *Mandates:*

- (1.1) — `graph` is an `adjacency_list`, which may be directed or undirected.

2 *Preconditions:*

- (2.1) —  $0 \leq \text{source} < \text{num\_vertices}(\text{graph})$ .
- (2.2) — `distance[i] == std::numeric_limits<range_value_t<D>>::max()` for  $0 \leq i < \text{num\_vertices}(\text{graph})$ . [ANDREW: invalid distance?]
- (2.3) — The `predecessors` range must be initialized so that `predecessors[i] == i` for all `i` such that  $0 \leq i < \text{num\_vertices}(\text{graph})$ .

3 *Effects:* Compute the breadth-first path and associated distance from vertex `source` to all reachable vertices in `graph`.

4 *Result:*

- (4.1) — If vertex with index `i` is reachable from vertex `source`, then `distances[i]` will contain the distance from `source` to vertex `i`. Otherwise, `distances[i]` will contain `std::numeric_limits<range_value_t<D>>::max()`.
- (4.2) — For `breadth_first_search` if vertex with index `i` is reachable from vertex `source`, then `predecessor[i]` will contain the predecessor vertex of vertex `i`. Otherwise, `predecessors[i]` will contain `i`.

5 *Returns:* `void`

6 *Throws:* none. [ANDREW: Throw if `source` out of range?]

7 *Complexity:*  $\mathcal{O}(|E| + |V|)$

8 *Remarks:*

### 3.3.2 Weighted Shortest Paths

#### 3.3.2.1 Dijkstra Initialization

[ANDREW: Actually, I don't think `dijkstra_invalid_distance()` or `dijkstra_zero()` useful – those need to be specified for the actual init functions. As free functions they don't really do anything. Or are they meant to be CPOs? ]

[ANDREW: Is there a run-time overhead introduced by using functions and/or CPOs for things like `dijkstra_invalid` or `dijkstra_zero`? ]

[ANDREW: I would like to see a use case for the `dijkstra_invalid` and `dijkstra_zero` functions.]

```
template <class DistanceValue>
auto dijkstra_invalid_distance() {
    return std::numeric_limits<DistanceValue>::max();
}

template <class DistanceValue>
auto dijkstra_zero() {
    return {};
}

template <property P, property D>
void init_dijkstra_shortest_paths(P&& predecessors, D&& distances);

template <property P, property D>
void init_dijkstra_shortest_paths(P&& predecessors,
    D&& distances,
    std::ranges::range_value_t<D> init);

template <property D>
```



```
void init_dijkstra_shortest_distances(D&& distances);

template <property D>
void init_dijkstra_shortest_distances(D&& distances, std::ranges::range_value_t<D> init);
```

1 *Effects:* :

- (1.1) — Each `predessors[i]` is initialized to `i`.
- (1.2) — Each `distance[i]` is initialized to `std::numeric_limits::max()` or to `init`.

### 3.3.2.2 Dijkstra Single Source Shortest Paths

Compute the shortest path and associated distance from vertex `source` to all reachable vertices in `graph` using non-negative weights.

|                                                          |                                             |                                           |
|----------------------------------------------------------|---------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}(( E  +  V ) \log  V )$ | <b>Throws? Yes</b><br><b>Multi-edge? No</b> | <b>Cycles? No</b><br><b>Directed? Yes</b> |
|----------------------------------------------------------|---------------------------------------------|-------------------------------------------|

Note that complexity may be  $\mathcal{O}(|E| + |V| \log |V|)$  for certain implementations.

```
// Concepts and types from std::ranges don't include the namespace prefix for brevity
// and clarity of purpose

// A fake range with no values and does nothing
inline static null_range_type null_predecessors;

template <index_adjacency_list G,
         random_access_range Distance,
         random_access_range Predecessor
         class WF = function<range_value_t<Distance>(edge_reference_t<G>)>>
requires convertible_to<vertex_id_t<G>, range_value_t<Predecessor>> &&
         edge_weight_function<G, WF, range_value_t<Distance>>
void dijkstra_shortest_paths(
    const G& graph,
    vertex_id_t<G> source,
    Distance& distances,
    Predecessor& predecessors = null_predecessors,
    WF&& w = [] (edge_reference_t<G> uv) { return range_value_t<Distance>(1); });

template <index_adjacency_list G,
         random_access_range Distance,
         random_access_range Predecessor,
         class Compare,
         class Combine,
         class WF = function<range_value_t<Distance>(edge_reference_t<G>)>>
requires convertible_to<vertex_id_t<G>, range_value_t<Predecessor>> &&
         basic_edge_weight_function<G, WF, range_value_t<Distance>, Compare, Combine>
void dijkstra_shortest_paths(
    const G& graph,
    vertex_id_t<G> source,
    Compare&& compare,
    Combine&& combine,
    Distance& distances,
    Predecessor& predecessors = null_predecessors,
    WF&& w = [] (edge_reference_t<G> uv) { return range_value_t<Distance>(1); });
```

1 *Mandates:*

- (1.1) — The weight function `w` must return a non-negative value.

2 *Preconditions:*

- (2.1) —  $0 \leq \text{source} < \text{num\_vertices}(\text{graph})$ .

- (2.2) — `distance[i] = numeric_limits<range_value_t<Distance>>::max()` for  $0 \leq i < \text{num\_vertices}(\text{graph})$ .  
 [ANDREW: invalid distance?]
- (2.3) — `predecessors[i] = i` for  $0 \leq i < \text{num\_vertices}(\text{graph})$ .
- 3 *Effects:*
- (3.1) — If vertex with index `i` is reachable from vertex `source`, then `distances[i]` will contain the distance from `source` to vertex `i`. Otherwise `distances[i]` will contain `numeric_limits<range_value_t<Distance>>::max()`.
- (3.2) — If vertex with index `i` is reachable from vertex `source`, then `predecessors[i]` will contain the predecessor vertex of vertex `i`. Otherwise `predecessors[i]` will contain `i`.
- 4 *Throws:* `out_of_range` is thrown when `source` is not in the range  $0 \leq \text{source} < \text{num\_vertices}(\text{graph})$ .
- 5 *Remarks:*
- (5.1) — Passing `null_predecessors` in the `predecessors` parameter will result in a compile-time decision to not evaluate predecessors.
- (5.2) — Bellman-Ford Shortest Paths allows negative weights with the consequence of greater complexity.

### 3.3.2.3 Bellman-Ford Single Source Shortest Paths

Compute the shortest path and associated distance from vertex `source` to all reachable vertices in `graph`.

|                                                   |                                             |                                           |
|---------------------------------------------------|---------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  \cdot  V )$ | <b>Throws? Yes</b><br><b>Multi-edge? No</b> | <b>Cycles? No</b><br><b>Directed? Yes</b> |
|---------------------------------------------------|---------------------------------------------|-------------------------------------------|

*// Concepts and types from std::ranges don't include the namespace prefix for brevity  
 // and clarity of purpose*

```
template <index_adjacency_list G,
         random_access_range Distance,
         random_access_range Predecessor
         class WF = function<range_value_t<Distance>(edge_reference_t<G>)>>
requires convertible_to<vertex_id_t<G>, range_value_t<Predecessor>> &&
         edge_weight_function<G, WF, range_value_t<Distance>>
void bellman_ford_shortest_paths(
    const G& graph,
    vertex_id_t<G> source,
    Distance& distances,
    Predecessor& predecessors = null_predecessors,
    WF&& w = [] (edge_reference_t<G> uv) { return range_value_t<Distance>(1); });

template <index_adjacency_list G,
         random_access_range Distance,
         random_access_range Predecessor,
         class Compare,
         class Combine,
         class WF = function<range_value_t<Distance>(edge_reference_t<G>)>>
requires convertible_to<vertex_id_t<G>, range_value_t<Predecessor>> &&
         basic_edge_weight_function<G, WF, range_value_t<Distance>, Compare, Combine>
void bellman_ford_shortest_paths(
    const G& graph,
    vertex_id_t<G> source,
    Compare&& compare,
    Combine&& combine,
    Distance& distances,
    Predecessor& predecessors = null_predecessors,
    WF&& w = [] (edge_reference_t<G> uv) { return range_value_t<Distance>(1); });
```

[PHIL: Should negative weight cycles be a pre-condition, or should it be detected with an exception thrown when it exists?]

- 1 *Preconditions:*
- (1.1) — `0 <= source < num_vertices(g).`
- (1.2) — `distance[i] = numeric_limits<range_value_t<Distance>>::max()` for `0 <= i < num_vertices(g).`  
[ANDREW: invalid distance?]
- (1.3) — `predecessors[i] = i` for `0 <= i < num_vertices(g).`
- 2 *Effects:*
- (2.1) — If vertex with index `i` is reachable from vertex `source`, then `distances[i]` will contain the distance from `source` to vertex `i`. Otherwise `distances[i]` will contain `numeric_limits<range_value_t<Distance>>::max()`.
- (2.2) — If vertex with index `i` is reachable from vertex `source`, then `predecessors[i]` will contain the predecessor vertex of vertex `i`. Otherwise `predecessors[i]` will contain `i`.
- 3 *Throws:* `out_of_range` is thrown when `source` is not in the range `0 <= source < num_vertices(graph)`.
- 4 *Remarks:*
- (4.1) — Passing `null_predecessors` in the `predecessors` parameter will result in a compile-time decision to not evaluate predecessors.
- (4.2) — Unlike Dijkstra's algorithm, Bellman-Ford allows negative edge weights. Performance constraints limit this to smaller graphs.

## 3.4 Clustering

### 3.4.1 Triangle Counting

[PHIL: Description needed]

|                                         |                                            |                                           |
|-----------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}(N^3)$ | <b>Throws? No</b><br><b>Multi-edge? No</b> | <b>Cycles? No</b><br><b>Directed? Yes</b> |
|-----------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
size_t triangle_count(G&& g) {}
```

- 1 *Returns:* Number of triangles

## 3.5 Communities

### 3.5.1 Label Propagation

[PHIL: Description needed]

|                                                    |                                            |                                           |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}(N)$ [KEVIN: ???] | <b>Throws? No</b><br><b>Multi-edge? No</b> | <b>Cycles? No</b><br><b>Directed? Yes</b> |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
         ranges::random_access_range Label,
         class Gen = std::default_random_engine,
         class T = size_t>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void label_propagation(G&& g,
                      Label& label,
                      Gen&& rng = std::default_random_engine {},
                      T max_iters = std::numeric_limits<T>::max()) {}
```

*Preconditions:*

- (1.1) — `label` contains vertex labels.
- (1.2) — `rng` is a random number generator.
- (1.3) — `max_iters` is the maximum number of iterations of the label propagation, or equivalently the maximum distance a label will propagate from its starting vertex.

*Effects:* `label[uid]` is the label assignments of vertex id `uid` discovered by label propagation.

|                                       |                                            |                                           |
|---------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}(N)$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|---------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
         ranges::random_access_range Label,
         class Gen = std::default_random_engine,
         class T = size_t>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void label_propagation(G&& g,
                      Label& label,
                      std::ranges::range_value_t<Label>& empty_label,
                      Gen&& rng = std::default_random_engine {},
                      T max_iters = std::numeric_limits<T>::max()) {}
```

*Preconditions:*

- (3.1) — `label` contains vertex labels.
- (3.2) — `empty_label` defines a label that is considered empty and will not be propagated.
- (3.3) — `rng` is a random number generator.
- (3.4) — `max_iters` is the maximum number of iterations of the label propagation, or equivalently the maximum distance a label will propagate from its starting vertex.

*Effects:* `label[uid]` is the label assignments of vertex id `uid` discovered by label propagation.

## 3.6 Components

### 3.6.1 Articulation Points

[PHIL: Description needed]

|                                               |                                            |                                           |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
         class Iter>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>> &&
         std::output_iterator<Iter, vertex_id_t<G>>
void articulation_points(G&& g,
                       Iter cut_vertices) {}
```

*Preconditions:*

- (1.1) — Output iterator `cut_vertices` can be assigned vertices of type `vertex_id_t<G>` when dereferenced.

*Effects:*

- (2.1) — Output iterator `cut_vertices` contains articulation point vertices, those which removed increase the number of components of `g`.

### 3.6.2 BiConnected Components

[PHIL: Description needed]

|                                               |                                            |                                           |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
          std::ranges::forward_range OuterContainer>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>> &&
          std::ranges::forward_range<std::ranges::range_value_t<OuterContainer>> &&
          integral<std::ranges::forward_range_t<std::ranges::forward_range_t<OuterContainer>>>
void biconnected_components(G&& g,
                           OuterContainer& components) {}
```

1 *Preconditions:*

- (1.1) — `components` is a container of containers. The inner container stores vertex ids.

2 *Effects:*

- (2.1) — `components` contains groups of biconnected components.

### 3.6.3 Connected Components

[PHIL: Description needed]

|                                               |                                            |                                          |
|-----------------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> No |
|-----------------------------------------------|--------------------------------------------|------------------------------------------|

```
template <adjacency_list G,
          ranges::random_access_range Component>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void connected_components(G&& g,
                         Component& component) {}
```

1 *Preconditions:*

- (1.1) — Size of `component` is greater than or equal to `num_vertices(g)`.

2 *Effects:*

- (2.1) — `component[v]` is the connected component id of `v`.

### 3.6.4 Strongly Connected Components

#### 3.6.4.1 Kosaraju's SCC

[PHIL: Description needed]

|                                               |                                            |                                           |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
          adjacency_list GT,
          ranges::random_access_range Component>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>> &&
          ranges::random_access_range<vertex_range_t<GT>> && integral<vertex_id_t<GT>>
void strongly_connected_components(G&& g,
                                  GT&& g_t,
                                  Component& component) {}
```

1 *Preconditions:*

- (1.1) — `g_t` is the transpose of `g`. Edge `uv` in `g` implies edge `vu` in `g_t`. `num_vertices(g)` equals `num_vertices(g_t)`.

- (1.2) — Size of `component` is greater than or equal to `num_vertices(g)`.

2 *Effects:*

- (2.1) — `component[v]` is the strongly connected component id of `v`.

### 3.6.4.2 Tarjan's SCC

[PHIL: Description needed]

|                                               |                                            |                                           |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E  +  V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G,
          ranges::random_access_range Component>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void strongly_connected_components(G&& g,
                                   Component& component) {}
```

1 *Preconditions:*

- (1.1) — Size of `component` is greater than or equal to `num_vertices(g)`.

2 *Effects:*

- (2.1) — `component[v]` is the strongly connected component id of `v`.

## 3.7 Directed Acyclic Graphs

### 3.7.1 Topological Sort

[PHIL: Description needed]

[PHIL: Do we need this as an algorithm, in addition to the view? What value does it bring?]

## 3.8 Maximal Independent Set

### 3.8.1 Maximal Independent Set

|                                         |                                            |                                          |
|-----------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> No |
|-----------------------------------------|--------------------------------------------|------------------------------------------|

```
template <adjacency_list G, class Iter>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>> &&
         std::output_iterator<Iter, vertex_id_t<G>>
void maximal_independent_set(G&& g, Iter mis, vertex_id_t<G> seed) {}
```

1 *Preconditions:*

- (1.1) —  $0 \leq \text{seed} < \text{num\_vertices}(\text{graph})$ .
- (1.2) — `mis` output iterator can be assigned vertices of type `vertex_id_t<G>` when dereferenced.

2 *Effects:*

- (2.1) — Output iterator `mis` contains maximal independent set of vertices containing `seed`, which is a subset of `vertices(graph)`.

## 3.9 Link Analysis

### 3.9.1 Jaccard Coefficient

[PHIL: Description needed]

|                                           |                                            |                                           |
|-------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( N ^3)$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-------------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <adjacency_list G, typename OutOp, typename T = double>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>> &&
    is_invocable_v<OutOp, vertex_id_t<G>&, vertex_id_t<G>&, edge_t<G>&, T>
void jaccard_coefficient(G&& g, OutOp out) {}
```

1 *Preconditions:*

- (1.1) — `out` is an operator for setting the resulting Jaccard coefficient. This function is expected to be of the form `out (vertex_id_t<G> uid, vertex_id_t<G> vid, edge_t<G> uv, T val)`.

2 *Effects:*

- (2.1) — For every pair of neighboring vertices (`uid`, `vid`), the function `out` is called, passing the vertex ids, the edge `uv` between them, and the calculated Jaccard coefficient.

## 3.10 Minimum Spanning Tree

### 3.10.1 Kruskal Minimum Spanning Tree

[PHIL: Description needed]

|                                         |                                            |                                           |
|-----------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> Yes |
|-----------------------------------------|--------------------------------------------|-------------------------------------------|

```
template <edgelist::edgelist E, edgelist::edgelist T>
void kruskal(E&& e, T&& t) {}

template <edgelist::edgelist E, edgelist::edgelist T, CompareOp>
void kruskal(E&& e, T&& t, CompareOp compare) {}
```

1 *Preconditions:*

- (1.1) — `e` is an `edgelist`.
- (1.2) — `compare` operator is a valid comparison operation on two edge values of type `edge_value_t<EL>` which returns a bool.

2 *Effects:*

- (2.1) — Edgelist `t` contains edges representing a spanning tree or forest, which minimize the comparison operator. When `compare` is `<`, `t` represents a minimum weight spanning tree.

### 3.10.2 Prim Minimum Spanning Tree

[PHIL: Description needed]

[PHIL: Use general form of dijkstra's shortest path?]

|                                                |                                            |                                          |
|------------------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Complexity</b><br>$\mathcal{O}( E \log V )$ | <b>Throws?</b> No<br><b>Multi-edge?</b> No | <b>Cycles?</b> No<br><b>Directed?</b> No |
|------------------------------------------------|--------------------------------------------|------------------------------------------|

```
template <adjacency_list G,
    ranges::random_access_range Predecessor,
    ranges::random_access_range Weight>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void prim(G&& g, Predecessor& predecessor, Weight& weight, vertex_id_t<G> seed = 0) {}

template <adjacency_list G,
    ranges::random_access_range Predecessor,
```

```

    ranges::random_access_range Weight,
    class CompareOp>
requires ranges::random_access_range<vertex_range_t<G>> && integral<vertex_id_t<G>>
void prim(G&& g,
    Predecessor& predecessor,
    Weight& weight,
    CompareOp compare,
    ranges::range_value_t<Weight> init_dist,
    vertex_id_t<G> seed = 0) {}

```

1 *Preconditions:*

- (1.1) — `0 <= seed < num_vertices(g)`.
- (1.2) — Size of `weight` and `predecessor` is greater than or equal to `num_vertices(g)`.
- (1.3) — `compare` operator is a valid comparison operation on two edge values of type `edge_value_t<G>` which returns a `bool`.

2 *Effects:*

- (2.1) — `predecessor[v]` is the parent vertex of `v` in a tree rooted at `seed` and `weight[v]` is the value of the edge between `v` and `predecessor[v]` in the tree. When `compare` is `<` and `init_dist==+inf`, `predecessor` represents a minimum weight spanning tree.
- (2.2) — If `predecessor` and `weight` are not initialized by user, and the graph is not fully connected, `predecessor[v]` and `weight[v]` will be undefined for vertices not in the same connected component as `seed`.



## Chapter4 Other Algorithms

Additional algorithms that were considered but not included in this proposal are identified in Table 4.1. It is assumed that future proposals will include them, with a recommendation of each Tier being in its own proposal. Tier X algorithms are variations of shortest paths algorithms that complement the Single Source, Multiple Target algorithms in this proposal.

| Tier 2                                                                                                                                                                                                                                                                | Tier 3                                                                     | Tier X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All Pairs Shortest Paths<br>Floyd-Warshall<br>Johnson<br>Centrality: Betweenness Centrality<br>Coloring: Greedy<br>Communities: Louvain<br>Connectivity: Minimum Cuts<br>Transitive Closure<br>Flows: Edmonds Karp<br>Flows: Push Relabel<br>Flows: Boykov Kolmogorov | Jones Plassman<br>Cores: k-cores<br>Cores: k-truss<br>Subgraph Isomorphism | Single Source, Single Target: Shortest Paths<br>Single Source, Single Target: BFS<br>Single Source, Single Target: Dijkstra<br>Single Source, Single Target: Bellman-Ford<br>Single Source, Single Target: Delta Stepping<br><br>Multiple Source: Shortest Paths<br>Multiple Source: BFS<br>Multiple Source: Dijkstra<br>Multiple Source: Bellman-Ford<br>Multiple Source: Delta Stepping<br><br>Multiple Source, Single Target: Shortest Paths<br>Multiple Source, Single Target: BFS<br>Multiple Source, Single Target: Dijkstra<br>Multiple Source, Single Target: Bellman-Ford<br>Multiple Source, Single Target: Delta Stepping |

Table 4.1 — Other Algorithms

[ANDREW: All Pairs: Tier 2? People bring this up alot – but it is very expensive in terms of computation and memory.]

# Chapter5 Views

The views in this section provide common ways that algorithms use to traverse graphs. They are as simple as iterating through the set of vertices, or more complex ways such as depth-first search and breadth-first search. They also provide a consistent and reliable way to access related elements using the View Return Types, and guaranteeing expected values, such as that the target is really the target on unordered edges.

## 5.1 Return Types (Descriptors)

Views return one of the types in this section, providing a consistent set of values. They are templated so that the view can adjust the actual values returned to be appropriate for its use. The three types, `vertex_descriptor`, `edge_descriptor` and `neighbor_descriptor`, define the data model used by the algorithms.

The following examples show the general design and how it's used. While it focuses on `vertexlist` to iterate over all vertices, it applies to all descriptors and view functions.

```
// the type of uu is vertex_descriptor<vertex_id_t<G>, vertex_reference_t<G>, void>
for(auto&& uu : vertexlist(g)) {
    vertex_id_t<G> id = uu.id;
    vertex_reference_t<G> u = uu.vertex;
    // ... do something interesting
}
```

Structured bindings make it simpler.

```
for(auto&& [id, u] : vertexlist(g)) {
    // ... do something interesting
}
```

A function object can also be passed to return a value from the vertex. In this case, `vertexlist(g)` returns `vertex_descriptor<vertex_id_t<G>, vertex_reference_t<G>, decltype(vvf(u))>`.

```
// the type returned by vertexlist is
// vertex_descriptor<vertex_id_t<G>,
// vertex_reference_t<G>,
// decltype(vvf(vertex_reference_t<G>))>
auto vvf = [&g](vertex_reference_t<G> u) { return vertex_value(g,u); };
for(auto&& [id, u, value] : vertexlist(g, vvf)) {
    // ... do something interesting
}
```

A simpler version also exists if all you need is a vertex id. The vertex value function takes a vertex id instead of a vertex reference.

```
for(auto&& [uid] : basic_vertexlist(g)) {
    // ... do something interesting
}

auto vvf = [&g](vertex_id_t<G> uid) { return vertex_value(g,uid); };
for(auto&& [uid, value] : basic_vertexlist(g,vvf)) {
    // ... do something interesting
}
```

### 5.1.0.1 struct vertex\_descriptor<VId, V, VV>

`vertex_descriptor` is used to return vertex information. It is used by `vertexlist(g)`, `vertices_breadth_first_search(g,u)`, `vertices_depth_first_search(g,u)` and others. The `id` member always exists.

```
template <class VId, class V, class VV>
struct vertex_descriptor {
    using id_type = VId; // e.g. vertex_id_t<G>
    using vertex_type = V; // e.g. vertex_reference_t<G> or void
    using value_type = VV; // e.g. vertex_value_t<G> or void

    id_type id;
    vertex_type vertex;
    value_type value;
};
```

Specializations are defined with `V=void` or `VV=void` to suppress the existence of their associated member variables, giving the following valid combinations in Table 5.1. For instance, the second entry, `vertex_descriptor<VId, V>` has two members `{id_type id; vertex_type vertex;}` and `value_type` is `void`.

| Template Arguments                                    | Members                                                |
|-------------------------------------------------------|--------------------------------------------------------|
| <code>vertex_descriptor&lt;VId, V, VV&gt;</code>      | <code>id</code> <code>vertex</code> <code>value</code> |
| <code>vertex_descriptor&lt;VId, V, void&gt;</code>    | <code>id</code> <code>vertex</code>                    |
| <code>vertex_descriptor&lt;VId, void, VV&gt;</code>   | <code>id</code> <code>value</code>                     |
| <code>vertex_descriptor&lt;VId, void, void&gt;</code> | <code>id</code>                                        |

Table 5.1 — `vertex_descriptor` Members

A useful type alias for copying vertex values (excluding the vertex reference) is also available. [PHIL: Duplicate; remove]

```
template <class VId, class VV>
using copyable_vertex_t = vertex_descriptor<VId, void, VV>; // id, value
```

#### 5.1.0.2 struct edge\_descriptor<VId, Sourced, E, EV>

`edge_descriptor` is used to return edge information. It is used by `incidence(g,u)`, `edgelist(g)`, `edges_breadth_first_search(g,u)`, `edges_depth_first_search(g,u)` and others. When `Sourced=true`, the `source_id` member is included with type `VId`. The `target_id` member always exists.

```
template <class VId, bool Sourced, class E, class EV>
struct edge_descriptor {
    using source_id_type = VId; // e.g. vertex_id_t<G> when Sourced==true, or void
    using target_id_type = VId; // e.g. vertex_id_t<G>
    using edge_type = E; // e.g. edge_reference_t<G> or void
    using value_type = EV; // e.g. edge_value_t<G> or void

    source_id_type source_id;
    target_id_type target_id;
    edge_type edge;
    value_type value;
};
```

Specializations are defined with `Sourced=true|false`, `E=void` or `EV=void` to suppress the existence of the associated member variables, giving the following valid combinations in Table 5.2. For instance, the second entry, `edge_descriptor<VId,true,E>` has three members `{source_id_type source_id; target_id_type target_id; edge_type edge;}` and `value_type` is `void`.

A useful type alias for copying edge values (excluding the edge reference) is also available. [PHIL: Duplicate; remove]

```
template <class VId, class EV>
using copyable_edge_t = edge_descriptor<VId, true, void, EV>; // source_id,target_id[,value]
```

| Template Arguments                                         | Members                |                        |                   |                    |
|------------------------------------------------------------|------------------------|------------------------|-------------------|--------------------|
| <code>edge_descriptor&lt;VId, true, E, EV&gt;</code>       | <code>source_id</code> | <code>target_id</code> | <code>edge</code> | <code>value</code> |
| <code>edge_descriptor&lt;VId, true, E, void&gt;</code>     | <code>source_id</code> | <code>target_id</code> | <code>edge</code> |                    |
| <code>edge_descriptor&lt;VId, true, void, EV&gt;</code>    | <code>source_id</code> | <code>target_id</code> |                   | <code>value</code> |
| <code>edge_descriptor&lt;VId, true, void, void&gt;</code>  | <code>source_id</code> | <code>target_id</code> |                   |                    |
| <code>edge_descriptor&lt;VId, false, E, EV&gt;</code>      |                        | <code>target_id</code> | <code>edge</code> | <code>value</code> |
| <code>edge_descriptor&lt;VId, false, E, void&gt;</code>    |                        | <code>target_id</code> | <code>edge</code> |                    |
| <code>edge_descriptor&lt;VId, false, void, EV&gt;</code>   |                        | <code>target_id</code> |                   | <code>value</code> |
| <code>edge_descriptor&lt;VId, false, void, void&gt;</code> |                        | <code>target_id</code> |                   |                    |

Table 5.2 — `edge_descriptor` Members

### 5.1.0.3 `struct neighbor_descriptor<VId, Sourced, V, VV>`

`neighbor_descriptor` is used to return information for a neighbor vertex, through an edge. It is used by `neighbors(g, u)`. When `Sourced=true`, the `source_id` member is included with type `source_id_type`. The `target_id` member always exists.

```
template <class VId, bool Sourced, class V, class VV>
struct neighbor_descriptor {
    using source_id_type = VId; // e.g. vertex_id_t<G> when Sourced==true, or void
    using target_id_type = VId; // e.g. vertex_id_t<G>
    using vertex_type = V; // e.g. vertex_reference_t<G> or void
    using value_type = VV; // e.g. vertex_value_t<G> or void

    source_id_type source_id;
    target_id_type target_id;
    vertex_type target;
    value_type value;
};
```

Specializations are defined with `Sourced=true|false`, `E=void` or `EV=void` to suppress the existence of the associated member variables, giving the following valid combinations in Table 5.3. For instance, the second entry, `neighbor_descriptor<VId, true, E>` has three members {`source_id_type source_id`; `target_id_type target_id`; `vertex_type target`; } and `value_type` is `void`.

| Template Arguments                                             | Members                |                        |                     |                    |
|----------------------------------------------------------------|------------------------|------------------------|---------------------|--------------------|
| <code>neighbor_descriptor&lt;VId, true, E, EV&gt;</code>       | <code>source_id</code> | <code>target_id</code> | <code>target</code> | <code>value</code> |
| <code>neighbor_descriptor&lt;VId, true, E, void&gt;</code>     | <code>source_id</code> | <code>target_id</code> | <code>target</code> |                    |
| <code>neighbor_descriptor&lt;VId, true, void, EV&gt;</code>    | <code>source_id</code> | <code>target_id</code> |                     | <code>value</code> |
| <code>neighbor_descriptor&lt;VId, true, void, void&gt;</code>  | <code>source_id</code> | <code>target_id</code> |                     |                    |
| <code>neighbor_descriptor&lt;VId, false, E, EV&gt;</code>      |                        | <code>target_id</code> | <code>target</code> | <code>value</code> |
| <code>neighbor_descriptor&lt;VId, false, E, void&gt;</code>    |                        | <code>target_id</code> | <code>target</code> |                    |
| <code>neighbor_descriptor&lt;VId, false, void, EV&gt;</code>   |                        | <code>target_id</code> |                     | <code>value</code> |
| <code>neighbor_descriptor&lt;VId, false, void, void&gt;</code> |                        | <code>target_id</code> |                     |                    |

Table 5.3 — `neighbor_descriptor` Members

## 5.2 Copyable Descriptors

### 5.2.1 Copyable Descriptor Types

Copyable descriptors are specializations of the descriptors that can be copied. More specifically, they don't include a vertex or edge reference. `copyable_vertex_t<G>` shows the simple definition.

```
template <class VId, class VV>
```

```
using copyable_vertex_t = vertex_descriptor<VId, void, VV>; // id, value
```

| Type                                             | Definition                                                  |
|--------------------------------------------------|-------------------------------------------------------------|
| <code>copyable_vertex_t&lt;T, VId, VV&gt;</code> | <code>vertex_descriptor&lt;VId, void, VV&gt;</code>         |
| <code>copyable_edge_t&lt;T, VId, EV&gt;</code>   | <code>edge_descriptor&lt;VId, true, void, EV&gt;&gt;</code> |
| <code>copyable_neighbor_t&lt;VId, VV&gt;</code>  | <code>neighbor_descriptor&lt;VId, true, void, VV&gt;</code> |

Table 5.4 — Descriptor Concepts

## 5.2.2 Copyable Descriptor Concepts

Given the copyable types, it's useful to have concepts to determine if a type is a desired copyable type.

| Concept                                          | Definition                                                               |
|--------------------------------------------------|--------------------------------------------------------------------------|
| <code>copyable_vertex&lt;T, VId, VV&gt;</code>   | <code>convertible_to&lt;T, copyable_vertex_t&lt;VId, VV&gt;&gt;</code>   |
| <code>copyable_edge&lt;T, VId, EV&gt;</code>     | <code>convertible_to&lt;T, copyable_edge_t&lt;VId, EV&gt;&gt;</code>     |
| <code>copyable_neighbor&lt;T, VId, VV&gt;</code> | <code>convertible_to&lt;T, copyable_neighbor_t&lt;VId, VV&gt;&gt;</code> |

Table 5.5 — Descriptor Concepts

## 5.3 Common Types and Functions for “Search”

[PHIL: Do these apply to all “search” functions?]

The `depth_first_search`, `breadth_first_search`, and `topological_sort` searches there are a number of common types and functions that apply to them.

Here are the types and functions for cancelling a search, getting the current depth of the search, and active elements in the search (e.g. number of vertices in a stack or queue).

```
// enum used to define how to cancel a search
enum struct cancel_search : int8_t {
    continue_search, // no change (ignored)
    cancel_branch, // stops searching from current vertex
    cancel_all // stops searching and dfs will be at end()
};

// stop searching from current vertex
template<class S>
void cancel(S search, cancel_search);

// Returns distance from the seed vertex to the current vertex,
// or to the target vertex for edge views
template<class S>
auto depth(S search) -> integral;

// Returns number of pending vertices to process
template<class S>
auto size(S search) -> integral;
```

Of particular note, `size(dfs)` is typically the same as `depth(dfs)` and is simple to calculate. `breadth_first_search` requires extra bookkeeping to evaluate `depth(bfs)` and returns a different value than `size(bfs)`.

The following example shows how the functions could be used, using `dfs` for one of the `depth_first_search` views. The same functions can be used for all search views.

```
auto&& g = ...; // graph
auto&& dfs = vertices_depth_first_search(g, 0); // start with vertex.id=0
for(auto&& [vid, v] : dfs) {
```

```

// No need to search deeper?
if(depth(dfs) > 3) {
    cancel(dfs, cancel_search::cancel_branch);
    continue;
}

if(size(dfs) > 1000) {
    std::cout << "Big depth of " << size(dfs) << '\n';
}

// do useful things
}

```

## 5.4 vertexlist Views

`vertexlist` views iterate over a range of vertices, returning a `vertex_descriptor` on each iteration. Table 5.6 shows the `vertexlist` functions overloads and their return values. `first` and `last` are vertex iterators.

| Example                                                                         | Return                                              |
|---------------------------------------------------------------------------------|-----------------------------------------------------|
| <code>for(auto&amp;&amp; [uid,u] : vertexlist(g))</code>                        | <code>vertex_descriptor&lt;VId,V,void&gt;</code>    |
| <code>for(auto&amp;&amp; [uid,u,val] : vertexlist(g,vvf))</code>                | <code>vertex_descriptor&lt;VId,V,VV&gt;</code>      |
| <code>for(auto&amp;&amp; [uid,u] : vertexlist(g,first,last))</code>             | <code>vertex_descriptor&lt;VId,V,void&gt;</code>    |
| <code>for(auto&amp;&amp; [uid,u,val] : vertexlist(g,first,last,vvf))</code>     | <code>vertex_descriptor&lt;VId,V,VV&gt;</code>      |
| <code>for(auto&amp;&amp; [uid,u] : vertexlist(g,vr))</code>                     | <code>vertex_descriptor&lt;VId,V,void&gt;</code>    |
| <code>for(auto&amp;&amp; [uid,u,val] : vertexlist(g,vr,vvf))</code>             | <code>vertex_descriptor&lt;VId,V,VV&gt;</code>      |
| <code>for(auto&amp;&amp; [uid] : basic_vertexlist(g))</code>                    | <code>vertex_descriptor&lt;VId,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [uid,val] : basic_vertexlist(g,vvf))</code>            | <code>vertex_descriptor&lt;VId,void,VV&gt;</code>   |
| <code>for(auto&amp;&amp; [uid] : basic_vertexlist(g,first,last))</code>         | <code>vertex_descriptor&lt;VId,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [uid,val] : basic_vertexlist(g,first,last,vvf))</code> | <code>vertex_descriptor&lt;VId,void,VV&gt;</code>   |
| <code>for(auto&amp;&amp; [uid] : basic_vertexlist(g,vr))</code>                 | <code>vertex_descriptor&lt;VId,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [uid,val] : basic_vertexlist(g,vr,vvf))</code>         | <code>vertex_descriptor&lt;VId,void,VV&gt;</code>   |

Table 5.6 — `vertexlist` View Functions

## 5.5 incidence Views

`incidence` views iterate over a range of adjacent edges of a vertex, returning a `edge_descriptor` on each iteration. Table 5.7 shows the `incidence` function overloads and their return values.

Since the source vertex `u` is available when calling an `incidence` function, there's no need to include sourced versions of the function to include `source_id` in the output.

| Example                                                                 | Return                                                  |
|-------------------------------------------------------------------------|---------------------------------------------------------|
| <code>for(auto&amp;&amp; [vid,uv] : incidence(g,uid))</code>            | <code>edge_descriptor&lt;VId,false,E,void&gt;</code>    |
| <code>for(auto&amp;&amp; [vid,uv,val] : incidence(g,uid,evf))</code>    | <code>edge_descriptor&lt;VId,false,E,EV&gt;</code>      |
| <code>for(auto&amp;&amp; [vid] : basic_incidence(g,uid))</code>         | <code>edge_descriptor&lt;VId,false,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [vid,val] : basic_incidence(g,uid,evf))</code> | <code>edge_descriptor&lt;VId,false,void,EV&gt;</code>   |

Table 5.7 — `incidence` View Functions

## 5.6 neighbors Views

`neighbors` views iterate over a range of edges for a vertex, returning a `vertex_descriptor` of each neighboring target vertex on each iteration. Table 5.8 shows the `neighbors` function overloads and their return values.

Since the source vertex `u` is available when calling a `neighbors` function, there's no need to include sourced versions of the function to include `source_id` in the output.

| Example                                                                 | Return                                                      |
|-------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>for(auto&amp;&amp; [vid,v] : neighbors(g,uid))</code>             | <code>neighbor_descriptor&lt;VId,false,V,void&gt;</code>    |
| <code>for(auto&amp;&amp; [vid,v,val] : neighbors(g,uid,vvf))</code>     | <code>neighbor_descriptor&lt;VId,false,V,VV&gt;</code>      |
| <code>for(auto&amp;&amp; [vid] : basic_neighbors(g,uid))</code>         | <code>neighbor_descriptor&lt;VId,false,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [vid,val] : basic_neighbors(g,uid,vvf))</code> | <code>neighbor_descriptor&lt;VId,false,void,VV&gt;</code>   |

Table 5.8 — `neighbors` View Functions

## 5.7 edgelist Views

`edgelist` views iterate over all edges for all vertices, returning a `edge_descriptor` on each iteration. Table 5.9 shows the `edgelist` function overloads and their return values.

| Example                                                               | Return                                                 |
|-----------------------------------------------------------------------|--------------------------------------------------------|
| <code>for(auto&amp;&amp; [uid,vid,uv] : edgelist(g))</code>           | <code>edge_descriptor&lt;VId,true,E,void&gt;</code>    |
| <code>for(auto&amp;&amp; [uid,vid,uv,val] : edgelist(g,evf))</code>   | <code>edge_descriptor&lt;VId,true,E,EV&gt;</code>      |
| <code>for(auto&amp;&amp; [uid,uv] : basic_edgelist(g))</code>         | <code>edge_descriptor&lt;VId,true,void,void&gt;</code> |
| <code>for(auto&amp;&amp; [uid,uv,val] : basic_edgelist(g,evf))</code> | <code>edge_descriptor&lt;VId,true,void,EV&gt;</code>   |

Table 5.9 — `edgelist` View Functions

## 5.8 depth\_first\_search Views

`depth_first_search` views iterate over the vertices and edges from a given seed vertex, returning a `vertex_descriptor` or `edge_descriptor` on each iteration when it is first encountered, depending on the function used. Table 5.10 shows the functions and their return values.

While not shown in the examples, all functions have a final, optional allocator parameter that defaults to `std::allocator<bool>`. It is used for containers that are internal to the view. The `<bool>` argument has no particular meaning.

[PHIL: Need "basic" variants that don't include edge reference]

| Example                                                                                          | Return                                               |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <code>for(auto&amp;&amp; [vid,v] : vertices_depth_first_search(g,seed))</code>                   | <code>vertex_descriptor&lt;VId,V,void&gt;</code>     |
| <code>for(auto&amp;&amp; [vid,v,val] : vertices_depth_first_search(g,seed,vvf))</code>           | <code>vertex_descriptor&lt;VId,V,VV&gt;</code>       |
| <code>for(auto&amp;&amp; [vid,uv] : edges_depth_first_search(g,seed))</code>                     | <code>edge_descriptor&lt;VId,false,E,void&gt;</code> |
| <code>for(auto&amp;&amp; [vid,uv,val] : edges_depth_first_search(g,seed,evf))</code>             | <code>edge_descriptor&lt;VId,false,E,EV&gt;</code>   |
| <code>for(auto&amp;&amp; [uid,vid,uv] : sourced_edges_depth_first_search(g,seed))</code>         | <code>edge_descriptor&lt;VId,true,E,void&gt;</code>  |
| <code>for(auto&amp;&amp; [uid,vid,uv,val] : sourced_edges_depth_first_search(g,seed,evf))</code> | <code>edge_descriptor&lt;VId,true,E,EV&gt;</code>    |

Table 5.10 — `depth_first_search` View Functions

## 5.9 breadth\_first\_search Views

`breadth_first_search` views iterate over the vertices and edges from a given seed vertex, returning a `vertex_descriptor` or `edge_descriptor` on each iteration when it is first encountered, depending on the function used. Table 5.11 shows the functions and their return values.

While not shown in the examples, all functions have a final, optional allocator parameter that defaults to `std::allocator<bool>`. It is used for containers that are internal to the view. The `<bool>` argument has no particular meaning.

[PHIL: Need "basic" variants that don't include edge reference]

## 5.10 topological\_sort Views

`topological_sort` views iterate over the vertices and edges from a given seed vertex, returning a `vertex_descriptor` or `edge_descriptor` on each iteration when it is first encountered, depending on the function used. Table 5.12 shows the functions and their return values.

| Example                                                                                            | Return                                               |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <code>for(auto&amp;&amp; [vid,v] : vertices_breadth_first_search(g,seed))</code>                   | <code>vertex_descriptor&lt;Vid,V,void&gt;</code>     |
| <code>for(auto&amp;&amp; [vid,v,val] : vertices_breadth_first_search(g,seed,vvf))</code>           | <code>vertex_descriptor&lt;Vid,V,VV&gt;</code>       |
| <code>for(auto&amp;&amp; [vid,uv] : edges_breadth_first_search(g,seed))</code>                     | <code>edge_descriptor&lt;Vid,false,E,void&gt;</code> |
| <code>for(auto&amp;&amp; [vid,uv,val] : edges_breadth_first_search(g,seed,evf))</code>             | <code>edge_descriptor&lt;Vid,false,E,EV&gt;</code>   |
| <code>for(auto&amp;&amp; [uid,vid,uv] : sourced_edges_breadth_first_search(g,seed))</code>         | <code>edge_descriptor&lt;Vid,true,E,void&gt;</code>  |
| <code>for(auto&amp;&amp; [uid,vid,uv,val] : sourced_edges_breadth_first_search(g,seed,evf))</code> | <code>edge_descriptor&lt;Vid,true,E,EV&gt;</code>    |

Table 5.11 — breadth\_first\_search View Functions

While not shown in the examples, all functions have a final, optional allocator parameter that defaults to `std::allocator<bool>`. It is used for containers that are internal to the view. The `<bool>` argument has no particular meaning.

[PHIL: Need "basic" variants that don't include edge reference]

| Example                                                                                        | Return                                               |
|------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <code>for(auto&amp;&amp; [vid,v] : vertices_topological_sort(g,seed))</code>                   | <code>vertex_descriptor&lt;Vid,V,void&gt;</code>     |
| <code>for(auto&amp;&amp; [vid,v,val] : vertices_topological_sort(g,seed,vvf))</code>           | <code>vertex_descriptor&lt;Vid,V,VV&gt;</code>       |
| <code>for(auto&amp;&amp; [vid,uv] : edges_topological_sort(g,seed))</code>                     | <code>edge_descriptor&lt;Vid,false,E,void&gt;</code> |
| <code>for(auto&amp;&amp; [vid,uv,val] : edges_topological_sort(g,seed,evf))</code>             | <code>edge_descriptor&lt;Vid,false,E,EV&gt;</code>   |
| <code>for(auto&amp;&amp; [uid,vid,uv] : sourced_edges_topological_sort(g,seed))</code>         | <code>edge_descriptor&lt;Vid,true,E,void&gt;</code>  |
| <code>for(auto&amp;&amp; [uid,vid,uv,val] : sourced_edges_topological_sort(g,seed,evf))</code> | <code>edge_descriptor&lt;Vid,true,E,EV&gt;</code>    |

Table 5.12 — topological\_sort View Functions

[PHIL: Is Topological Sort a view, an algorithm or both?]



# Chapter6 Graph Container Interface

The Graph Container Interface defines the primitive concepts, traits, types and functions used to define and access an adjacency graph, no matter its internal design and organization. Thus, it is designed to reflect all forms of adjacency graphs including a vector of lists, CSR-based graph and adjacency matrix, whether they are in the standard or external to the standard.

All algorithms in this proposal require that vertices are stored in random access containers and that `vertex_id_t<G>` is integral, and it is assumed that all future algorithm proposals will also have the same requirements.

The Graph Container Interface is designed to support a wider scope of graph containers than required by the views and algorithms in this proposal. This enables for future growth of the graph data model (e.g. incoming edges on a vertex), or as a framework for graph implementations outside of the standard. For instance, existing implementations may have requirements that cause them to define features with looser constraints, such as sparse vertex\_ids, non-integral vertex\_ids, or storing vertices in associative bi-directional containers (e.g. `std::map` or `std::unordered_map`). Such features require specialized implementations for views and algorithms. The performance for such algorithms will be sub-optimal, but is preferable to run them on the existing container rather than loading the graph into a high-performance graph container and then running the algorithm on it, where the loading time can far outweigh the time to run the sub-optimal algorithm. To achieve this, care has been taken to make sure that the use of concepts chosen is appropriate for algorithm, view and container.

## 6.1 Naming Conventions

Table 6.1 shows the naming conventions used throughout this document.

## 6.2 Concepts

Table 6.2 summarizes the concepts in the Graph Container Interface, allowing views and algorithms to verify a graph implementation has the expected requirements for an `adjacency_list` or `sourced_adjacency_list`.

Sourced edges have a `source_id` on them in addition to a `target_id`. A `sourced_adjacency_list` has sourced edges.

Indexed adjacency lists reflect a common use case where vertices are kept in a random access container and have an integral id.

[PHIL: Need to resolve how to deal with "basic" concepts which don't require vertex or edge references. Can a GGI exist without the references? Is it just a View artifact? ]

## 6.3 Traits

Table 6.3 summarizes the type traits in the Graph Container Interface, allowing views and algorithms to query the graph's characteristics.

## 6.4 Types

Table 6.4 summarizes the type aliases in the Graph Container Interface. These are the types used to define the objects in a graph container, no matter its internal design and organization. Thus, it is designed to be able to reflect all forms of adjacency graphs including a vector of lists, compressed\_graph and adjacency matrix.

The type aliases are defined by either a function specialization for the underlying graph container, or a refinement of one of those types (e.g. an iterator of a range). Table 6.5 describes the functions in more detail.

`graph_value(g)`, `vertex_value(g,u)` and `edge_value(g,uv)` can be optionally implemented, depending on whether the graph container supports values on the graph, vertex and edge types.

There is no contiguous requirement for `vertex_id` from one partition to the next, though in practice they will often be assigned contiguously. Gaps in `vertex_id`s between partitions should be allowed.

## 6.5 Classes and Structs

The `graph_error` exception class is available, inherited from `runtime_error`. While any function may use it, it is only anticipated to be used by the `load` functions at this time. No additional functionality is added beyond that provided by `runtime_error`.

| Template Parameter | Type Alias                                                                 | Variable Names                                  | Description                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G                  |                                                                            |                                                 | Graph                                                                                                                                                                                                                            |
|                    | <code>graph_reference_t&lt;G&gt;</code>                                    | <code>g</code>                                  | Graph reference                                                                                                                                                                                                                  |
| GV                 |                                                                            | <code>val</code>                                | Graph Value, value or reference                                                                                                                                                                                                  |
| V                  | <code>vertex_t&lt;G&gt;</code><br><code>vertex_reference_t&lt;G&gt;</code> | <code>u, v, x, y</code>                         | Vertex<br>Vertex reference. <code>u</code> is the source (or only) vertex. <code>v</code> is the target vertex.                                                                                                                  |
| VId                | <code>vertex_id_t&lt;G&gt;</code>                                          | <code>uid, vid, seed</code>                     | Vertex id. <code>uid</code> is the source (or only) vertex id. <code>vid</code> is the target vertex id.                                                                                                                         |
| VV                 | <code>vertex_value_t&lt;G&gt;</code>                                       | <code>val</code>                                | Vertex Value, value or reference. This can be either the user-defined value on a vertex, or a value returned by a function object (e.g. <code>VVF</code> ) that is related to the vertex.                                        |
| VR                 | <code>vertex_range_t&lt;G&gt;</code>                                       | <code>ur, vr</code>                             | Vertex Range                                                                                                                                                                                                                     |
| VI                 | <code>vertex_iterator_t&lt;G&gt;</code>                                    | <code>ui, vi</code><br><code>first, last</code> | Vertex Iterator. <code>ui</code> is the source (or only) vertex. <code>vi</code> is the target vertex.                                                                                                                           |
| VVF                |                                                                            | <code>vvf</code>                                | Vertex Value Function: <code>vvf(u) → vertex value</code>                                                                                                                                                                        |
| VProj              |                                                                            | <code>vproj</code>                              | Vertex descriptor projection function: <code>vproj(x) → vertex_descriptor&lt;VId, VV&gt;</code>                                                                                                                                  |
| <hr/>              |                                                                            |                                                 |                                                                                                                                                                                                                                  |
|                    | <code>partition_id_t&lt;G&gt;</code>                                       | <code>pid</code><br><code>P</code>              | Partition id.<br>Number of partitions.                                                                                                                                                                                           |
| PVR                | <code>partition_vertex_range_t&lt;G&gt;</code>                             | <code>pur, pvr</code>                           | Partition vertex range.                                                                                                                                                                                                          |
| E                  | <code>edge_t&lt;G&gt;</code><br><code>edge_reference_t&lt;G&gt;</code>     | <code>uv, vw</code>                             | Edge<br>Edge reference. <code>uv</code> is an edge from vertices <code>u</code> to <code>v</code> . <code>vw</code> is an edge from vertices <code>v</code> to <code>w</code> .                                                  |
| EV                 | <code>edge_value_t&lt;G&gt;</code>                                         | <code>val</code>                                | Edge Value, value or reference. This can be either the user-defined value on an edge, or a value returned by a function object (e.g. <code>EVF</code> ) that is related to the edge.                                             |
| ER                 | <code>vertex_edge_range_t&lt;G&gt;</code>                                  |                                                 | Edge Range for edges of a vertex                                                                                                                                                                                                 |
| EI                 | <code>vertex_edge_iterator_t&lt;G&gt;</code>                               | <code>uvi, vwi</code>                           | Edge Iterator for an edge of a vertex. <code>uvi</code> is an iterator for an edge from vertices <code>u</code> to <code>v</code> . <code>vwi</code> is an iterator for an edge from vertices <code>v</code> to <code>w</code> . |
| EVF                |                                                                            | <code>evf</code>                                | Edge Value Function: <code>evf(uv) → edge value</code>                                                                                                                                                                           |
| EProj              |                                                                            | <code>eproj</code>                              | Edge descriptor projection function: <code>eproj(x) → edge_descriptor&lt;VId, Sourced, EV&gt;</code>                                                                                                                             |
| <hr/>              |                                                                            |                                                 |                                                                                                                                                                                                                                  |
| PER                | <code>partition_edge_range_t&lt;G&gt;</code>                               |                                                 | Partition Edge Range for edges of a partition vertex.                                                                                                                                                                            |

Table 6.1 — Naming Conventions for Types and Variables

## 6.6 Functions

Table 6.5 summarizes the functions in the Graph Container Interface. These are the primitive functions used to access an adjacency graph, no matter its internal design and organization. Thus, it is designed to be able to reflect all forms of adjacency graphs including a vector of lists, CSR-based graph and adjacency matrix.

Functions that have n/a for their Default Implementation must be defined by the author of a Graph Container implementation.

Value functions (`graph_value(g)`, `vertex_value(g, u)` and `edge_value(g, uv)`) can be optionally implemented, depending on whether the graph container supports values on the graph, vertex and edge types. They return a single value and can be scalar, struct, class, union, or tuple. These are abstract types used by the GVF, VVF and EVF function objects to retrieve values used by algorithms. As such it's valid to return the "enclosing" owning class (graph, vertex or edge), or some other embedded value in those objects.

`vertex_id_t<G>` is defined by the type returned by `vertex_id(g)` and it defaults to the `difference_type` of the underlying container used for vertices (e.g. `int64_t` for 64-bit systems). This is sufficient for all situations. However, there are often space

| Concept                                      | Definition                                                                                                                                                      |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>vertex_range&lt;G&gt;</code>           | <code>vertices(g)</code> returns a sized, forward_range; <code>vertex_id(g, ui)</code> exists                                                                   |
| <code>targeted_edge&lt;G&gt;</code>          | <code>target_id(g, uv)</code> and <code>target(g, uv)</code> exist                                                                                              |
| <code>adjacency_list&lt;G&gt;</code>         | Extends <code>vertex_range&lt;G&gt;</code> by adding <code>edges(g, u)</code> and <code>edges(g, uid)</code> that returns a forward_range                       |
| <code>sourced_edge&lt;G&gt;</code>           | <code>source_id(g, uv)</code> and <code>source(g, uv)</code> exist                                                                                              |
| <code>sourced_adjacency_list&lt;G&gt;</code> | <code>adjacency_list&lt;G&gt;</code> and <code>sourced_edge&lt;G, edge_t&lt;G&gt;&gt;</code> and <code>edge_id(g, uv)</code> exists                             |
| <code>index_vertex_range&lt;G&gt;</code>     | Extends <code>vertex_range&lt;G&gt;</code> by requiring <code>vertices(g)</code> return a random_access_range and <code>vertex_id(g)</code> return an integer   |
| <code>index_adjacency_list&lt;G&gt;</code>   | Extends <code>adjacency_list&lt;G&gt;</code> by requiring <code>vertices(g)</code> return a random_access_range and <code>vertex_id(g)</code> return an integer |

Table 6.2 — Graph Container Interface Concepts

| Trait                                                       | Type       | Comment                                                                                                                    |
|-------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>has_degree&lt;G&gt;</code>                            | concept    | Is the <code>degree(g, u)</code> function available?                                                                       |
| <code>has_find_vertex&lt;G&gt;</code>                       | concept    | Are the <code>find_vertex(g, _)</code> functions available?                                                                |
| <code>has_find_vertex_edge&lt;G&gt;</code>                  | concept    | Are the <code>find_vertex_edge(g, _)</code> functions available?                                                           |
| <code>has_contains_edge&lt;G&gt;</code>                     | concept    | Is the <code>contains_edge(g, uid, vid)</code> function available?                                                         |
| <code>define_unordered_edge&lt;G, E&gt; : false_type</code> | struct     | Specialize for edge implementation to derive from <code>true_type</code> for unordered edges                               |
| <code>is_unordered_edge&lt;G, E&gt;</code>                  | struct     | <code>conjunction&lt;define_unordered_edge&lt;E&gt;, is_sourced_edge&lt;G, E&gt;&gt;</code>                                |
| <code>is_unordered_edge_v&lt;G, E&gt;</code>                | type alias |                                                                                                                            |
| <code>unordered_edge&lt;G, E&gt;</code>                     | concept    |                                                                                                                            |
| <code>is_ordered_edge&lt;G, E&gt;</code>                    | struct     | <code>negation&lt;is_unordered_edge&lt;G, E&gt;&gt;</code>                                                                 |
| <code>is_ordered_edge_v&lt;G, E&gt;</code>                  | type alias |                                                                                                                            |
| <code>ordered_edge&lt;G, E&gt;</code>                       | concept    |                                                                                                                            |
| <code>define_adjacency_matrix&lt;G&gt; : false_type</code>  | struct     | Specialize for graph implementation to derive from <code>true_type</code> for edges stored as a square 2-dimensional array |
| <code>is_adjacency_matrix&lt;G&gt;</code>                   | struct     |                                                                                                                            |
| <code>is_adjacency_matrix_v&lt;G&gt;</code>                 | type alias |                                                                                                                            |
| <code>adjacency_matrix&lt;G&gt;</code>                      | concept    |                                                                                                                            |

Table 6.3 — Graph Container Interface Type Traits

and performance advantages if a smaller type is used, such as `int32_t` or even `int16_t`. It is recommended to consider overriding this function for optimal results, assuring that it is also large enough for the number of possible vertices and edges in the application. It will also need to be overridden if the implementation doesn't expose the vertices as a range.

`find_vertex(g, uid)` is constant complexity because all algorithms in this proposal require that `vertex_range_t<G>` is a random access range.

If the concept requirements for the default implementation aren't met by the graph container the function will need to be overridden.

Edgelist are assumed to be either be an edgelist view of an adjacency graph, or a standard range with `source_id` and `target_id` values. There is no need for additional functions when a range is used.

## 6.7 Unipartite, Bipartite and Multipartite Graph Representation

`partition_count(g)` returns the number of partitions, or partiteness, of the graph. It has a range of 1 to `n`, where 1 identifies a unipartite graph, 2 is a bipartite graph, and a value of 2 or more can be considered a multipartite graph.

| Type Alias                                                                                                          | Definition                                                                 | Comment  |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------|
| <code>graph_reference_t&lt;G&gt;</code>                                                                             | <code>add_lvalue_reference&lt;G&gt;</code>                                 |          |
| <code>graph_value_t&lt;G&gt;</code>                                                                                 | <code>decltype (graph_value (g) )</code>                                   | optional |
| <code>vertex_range_t&lt;G&gt;</code>                                                                                | <code>decltype (vertices (g) )</code>                                      |          |
| <code>vertex_iterator_t&lt;G&gt;</code>                                                                             | <code>iterator_t&lt;vertex_range_t&lt;G&gt;&gt;</code>                     |          |
| <code>vertex_t&lt;G&gt;</code>                                                                                      | <code>range_value_t&lt;vertex_range_t&lt;G&gt;&gt;</code>                  |          |
| <code>vertex_reference_t&lt;G&gt;</code>                                                                            | <code>range_reference_t&lt;vertex_range_t&lt;G&gt;&gt;</code>              |          |
| <code>vertex_id_t&lt;G&gt;</code>                                                                                   | <code>decltype (vertex_id (g) )</code>                                     |          |
| <code>vertex_value_t&lt;G&gt;</code>                                                                                | <code>decltype (vertex_value (g) )</code>                                  | optional |
| <code>vertex_edge_range_t&lt;G&gt;</code>                                                                           | <code>decltype (edges (g, u) )</code>                                      |          |
| <code>vertex_edge_iterator_t&lt;G&gt;</code>                                                                        | <code>iterator_t&lt;vertex_edge_range_t&lt;G&gt;&gt;</code>                |          |
| <code>edge_t&lt;G&gt;</code>                                                                                        | <code>range_value_t&lt;vertex_edge_range_t&lt;G&gt;&gt;</code>             |          |
| <code>edge_reference_t&lt;G&gt;</code>                                                                              | <code>range_reference_t&lt;vertex_edge_range_t&lt;G&gt;&gt;</code>         |          |
| <code>edge_value_t&lt;G&gt;</code>                                                                                  | <code>decltype (edge_value (g) )</code>                                    | optional |
| ----- The following is only available when the optional <code>source_id(g, uv)</code> is defined for the edge ----- |                                                                            |          |
| <code>edge_id_t&lt;G&gt;</code>                                                                                     | <code>edge_descriptor&lt;vertex_id_t&lt;G&gt;, true, void, void&gt;</code> |          |
| <code>partition_id_t&lt;G&gt;</code>                                                                                | <code>decltype (partition_id (g, u) )</code>                               | optional |
| <code>partition_vertex_range_t&lt;G&gt;</code>                                                                      | <code>vertices (g, pid)</code>                                             | optional |
| <code>partition_edge_range_t&lt;G&gt;</code>                                                                        | <code>edges (g, u, pid)</code>                                             | optional |

Table 6.4 — Graph Container Interface Type Aliases

If a graph data structure doesn't support partitions then it is unipartite with one partition and partite functions will reflect that. For instance, `partition_count(g)` returns a value of 1, and `vertices(g, 0)` (vertices in the first partition) will return a range that includes all vertices in the graph.

A partition identifies a type of a vertex, where the vertex value types are assumed to be uniform in each partition. This creates a dilemma because the existing `vertex_value(g, u)` returns a single type based template parameter for the vertex value type. Supporting multiple types can be addressed in different ways using C++ features. The key to remember is that the actual value used by algorithms is done by calling a function object that retrieves the value to be used. That function is specific to the graph data structure, using the partition to determine how to get the appropriate value.

- `std::variant` : The lambda returns the appropriate variant value based on the partition.
- Base class pointer: The lambda can call a member function to return the value based on the partition.
- `void*` : The lambda can cast the pointer to a concrete type based on the partition, and then return the appropriate value.

`edges(g, uid, pid)` and `edges(g, ui, pid)` filter the edges where the target is in the partition `pid` passed. This isn't needed for bipartite graphs.

## 6.8 Loading Graph Data

The `load` functions are used to load vertex and edge data into a graph. They may throw a `graph_error` exception.

All graph data structures need to implement `load_graph`, `load_vertices` and `load_edges`. Whether `load_vertices` or `load_edges` can be called multiple times, or after `load_graph` is called, is dependent on the underlying graph data structure. `load_partition` only needs to be implemented if a graph supports partitions.

Projections are used to convert values in the input range to the expected copyable type. In the following `load_vertices` prototype, `vproj(ranges::range_value_t<VRng>&) → vertex_descriptor<vertex_id_t<G>, vertex_value_t<G>>`. If there is no vertex value stored in the graph then `vertex_value_t<G>` will be `void` and the resulting `vertex_descriptor` will have a single id member. If `vproj(ranges::range_value_t<VRng>&)` is the same as `vertex_descriptor<vertex_id_t<G>, vertex_value_t<G>>` then `VProj = identity` can be used.

```
template <adjacency_list G, ranges::forward_range VRng, class VProj = identity>
requires copyable_vertex<invoke_result<VProj, ranges::range_value_t<VRng>>,
    vertex_id_t<G>, vertex_value_t<G>>
constexpr void load_vertices(G&, const VRng& vrng, VProj vproj);
```

| Function                                                                                                 | Return Type                                    | Complexity | Default Implementation                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>graph_value(g)</code>                                                                              | <code>graph_value_t&lt;G&gt;</code>            | constant   | n/a, optional                                                                                                                                                         |
| <code>partition_count(g)</code>                                                                          | <code>vertex_id_t&lt;G&gt;</code>              | constant   | 1                                                                                                                                                                     |
| <code>vertices(g)</code>                                                                                 | <code>vertex_range_t&lt;G&gt;</code>           | constant   | <code>g</code> if <code>random_access_range&lt;G&gt;</code> , n/a otherwise                                                                                           |
| <code>num_vertices(g)</code>                                                                             | integral                                       | constant   | <code>size(vertices(g))</code>                                                                                                                                        |
| <code>find_vertex(g, uid)</code>                                                                         | <code>vertex_iterator_t&lt;G&gt;</code>        | constant   | <code>begin(vertices(g)) + uid</code><br>if <code>random_access_range&lt;vertex_range_t&lt;G&gt;&gt;</code>                                                           |
| <code>vertex_id(g, ui)</code>                                                                            | <code>vertex_id_t&lt;G&gt;</code>              | constant   | <code>ui - begin(vertices(g))</code><br>Override to define a different <code>vertex_id_t&lt;G&gt;</code> type (e.g. <code>int32_t</code> ).                           |
| <code>vertex_value(g, u)</code>                                                                          | <code>vertex_value_t&lt;G&gt;</code>           | constant   | n/a, optional                                                                                                                                                         |
| <code>vertex_value(g, uid)</code>                                                                        | <code>vertex_value_t&lt;G&gt;</code>           | constant   | <code>vertex_value(g, *find_vertex(g, uid))</code> , optional                                                                                                         |
| <code>degree(g, u)</code>                                                                                | integral                                       | constant   | <code>size(edges(g, u))</code> if <code>sized_range&lt;vertex_edge_range_t&lt;G&gt;&gt;</code>                                                                        |
| <code>degree(g, uid)</code>                                                                              | integral                                       | constant   | <code>size(edges(g, uid))</code> if <code>sized_range&lt;vertex_edge_range_t&lt;G&gt;&gt;</code>                                                                      |
| <code>partition_id(g, u)</code>                                                                          | <code>partition_id_t&lt;G&gt;</code>           | constant   | 0                                                                                                                                                                     |
| <code>partition_id(g, uid)</code>                                                                        | <code>partition_id_t&lt;G&gt;</code>           | constant   | <code>partition_id(g, *find_vertex(g, uid))</code>                                                                                                                    |
| <code>vertices(g, pid)</code>                                                                            | <code>partition_vertex_range_t&lt;G&gt;</code> | constant   | <code>vertices(g)</code>                                                                                                                                              |
| <code>num_vertices(g, pid)</code>                                                                        | integral                                       | constant   | <code>size(vertices(g))</code>                                                                                                                                        |
| <code>edges(g, u)</code>                                                                                 | <code>vertex_edge_range_t&lt;G&gt;</code>      | constant   | <code>u</code> if <code>forward_range&lt;vertex_t&lt;G&gt;&gt;</code> , n/a otherwise                                                                                 |
| <code>edges(g, uid)</code>                                                                               | <code>vertex_edge_range_t&lt;G&gt;</code>      | constant   | <code>edges(g, *find_vertex(g, uid))</code>                                                                                                                           |
| <code>target_id(g, uv)</code>                                                                            | <code>vertex_id_t&lt;G&gt;</code>              | constant   | n/a                                                                                                                                                                   |
| <code>target(g, uv)</code>                                                                               | <code>vertex_t&lt;G&gt;</code>                 | constant   | <code>* (begin(vertices(g)) + target_id(g, uv))</code> if <code>random_access_range&lt;vertex_range_t&lt;G&gt;&gt; &amp;&amp; integral&lt;target_id(g, uv)&gt;</code> |
| <code>edge_value(g, uv)</code>                                                                           | <code>edge_value_t&lt;G&gt;</code>             | constant   | <code>uv</code> if <code>forward_range&lt;vertex_t&lt;G&gt;&gt;</code> , n/a otherwise, optional                                                                      |
| <code>find_vertex_edge(g, u, vid)</code>                                                                 | <code>vertex_edge_t&lt;G&gt;</code>            | linear     | <code>find(edges(g, u), [] (uv) target_id(g, uv) == vid; )</code>                                                                                                     |
| <code>find_vertex_edge(g, uid, vid)</code>                                                               | <code>vertex_edge_t&lt;G&gt;</code>            | linear     | <code>find_vertex_edge(g, *find_vertex(g, uid), vid)</code>                                                                                                           |
| <code>contains_edge(g, uid, vid)</code>                                                                  | <code>bool</code>                              | constant   | <code>uid &lt; size(vertices(g)) &amp;&amp; vid &lt; size(vertices(g))</code> if <code>is_adjacency_matrix_v&lt;G&gt;</code> .                                        |
|                                                                                                          |                                                | linear     | <code>find_vertex_edge(g, uid) != end(edges(g, uid))</code> otherwise.                                                                                                |
| <code>edges(g, u, pid)</code>                                                                            | <code>partition_edge_range_t&lt;G&gt;</code>   | linear     | <code>edges(g, u)</code>                                                                                                                                              |
| <code>edges(g, uid, pid)</code>                                                                          | <code>partition_edge_range_t&lt;G&gt;</code>   | linear     | <code>edges(g, uid)</code>                                                                                                                                            |
| The following are only available when the optional <code>source_id(g, uv)</code> is defined for the edge |                                                |            |                                                                                                                                                                       |
| <code>source_id(g, uv)</code>                                                                            | <code>vertex_id_t&lt;G&gt;</code>              | constant   | n/a, optional                                                                                                                                                         |
| <code>source(g, uv)</code>                                                                               | <code>vertex_t&lt;G&gt;</code>                 | constant   | <code>* (begin(vertices(g)) + source_id(g, uv))</code> if <code>random_access_range&lt;vertex_range_t&lt;G&gt;&gt; &amp;&amp; integral&lt;target_id(g, uv)&gt;</code> |
| <code>edge_id(g, uv)</code>                                                                              | <code>edge_id_t&lt;G&gt;</code>                | constant   | <code>edge_descriptor&lt;vertex_id_t&lt;G&gt;, true, void, void&gt;{source_id(g, uv), target_id(g, uv)}</code>                                                        |

Table 6.5 — Graph Container Interface Functions

The same pattern is applied using `ERng` and `EProj` for edges.

For graphs with vertex values, `load_vertices` should be called before `load_edges`.

Whether `load_vertices` or `load_edges` can be called multiple times is graph-dependent.

For graphs with partitions, `load_partition` must be called to load vertices for each partition `pid`. `pid` values must be contiguous and their vertices should be loaded contiguously. `empty(vrng)` may be empty if there are no vertices in the partition.

| Function                                                                   | Return Type | Complexity | Default Implementation                                                                                  |
|----------------------------------------------------------------------------|-------------|------------|---------------------------------------------------------------------------------------------------------|
| <code>load_graph(g, erng, vrng, eproj=identity(), vproj=identity())</code> | void        | V + E      | n/a                                                                                                     |
| <code>load_vertices(g, vrng, vproj=identity())</code>                      | void        | V          | n/a                                                                                                     |
| <code>load_partition(g, pid, vrng, vproj=identity())</code>                | void        | V(p)       | <code>load_vertices</code> is called if partitions are not supported; there will be a single partition. |
| <code>load_edges(g, erng, eproj=identity(), vertex_count=0)</code>         | void        | E          | n/a                                                                                                     |

Table 6.6 — Graph Load Functions

## 6.9 Using Existing Graph Data Structures

Reasonable defaults have been defined for the GCI functions to minimize the amount of work needed to adapt an existing graph data structure to be used by the views and algorithms.

There are two cases supported. The first is for the use of standard containers to define the graph and the other is for a broader set of more complicated implementations.

### 6.9.1 Using Standard Containers for the Graph Data Structure

For example this we'll use `G = vector<forward_list<tuple<int, double>>>` to define the graph, where `g` is an instance of `G`. `tuple<int, double>` defines the `target_id` and `weight` property respectively. We can write loops to go through the vertices, and edges within each vertex, as follows.

```
using G = vector<forward_list<tuple<int, double>>>;
auto target_id(const G& g, edge_t<const G& uv) { return get<0>(uv); }
auto weight = [&g](edge_t& uv) { return get<1>(uv); }

G g;
load_graph(g, ...); // load some data

// Using GCI functions
for(auto&& [uid, u] : vertices(g)) {
    for(auto&& [vid, uv] : edges(g, u)) {
        auto w = weight(uv);
        // do something...
    }
}
```

Note that `target_id(g, uv)` was the only CPO function overridden; all other functions were automatically defined based on the rules shown in Table 6.7.

| Function or Value              | Concrete Type                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>vertices(g)</code>       | <code>vector&lt;forward_list&lt;tuple&lt;int, double&gt;&gt;&gt;</code> (when <code>random_access_range&lt;G&gt;</code> )         |
| <code>u</code>                 | <code>forward_list&lt;tuple&lt;int, double&gt;&gt;</code>                                                                         |
| <code>edges(g, u)</code>       | <code>forward_list&lt;tuple&lt;int, double&gt;&gt;</code> (when <code>random_access_range&lt;vertex_range_t&lt;G&gt;&gt;</code> ) |
| <code>uv</code>                | <code>tuple&lt;int, double&gt;</code>                                                                                             |
| <code>edge_value(g, uv)</code> | <code>tuple&lt;int, double&gt;</code> (when <code>random_access_range&lt;vertex_range_t&lt;G&gt;&gt;</code> )                     |

Table 6.7 — Types When Using Standard Containers

### 6.9.2 Using Other Graph Data Structures

For other graph data structures more function overrides are required. The following table identifies the common function overrides anticipated for most cases, keeping in mind that all functions in Table 6.5 can be overridden.

| Function                                                       | Comment                                |
|----------------------------------------------------------------|----------------------------------------|
| <code>vertices(g)</code>                                       |                                        |
| <code>edges(g, u)</code>                                       |                                        |
| <code>target_id(g, uv)</code>                                  |                                        |
| <code>edge_value(g, uv)</code>                                 | If edges have value(s) in the graph    |
| <code>vertex_value(g, u)</code>                                | If vertices have value(s) in the graph |
| <code>graph_value(g)</code>                                    | If the graph has value(s)              |
| When edges have the optional <code>source_id</code> on an edge |                                        |
| <code>source_id(g, uv)</code>                                  |                                        |
| When the graph supports multiple partitions                    |                                        |
| <code>partition_count(g)</code>                                |                                        |
| <code>partition_id(g, u)</code>                                |                                        |
| <code>vertices(g, u, pid)</code>                               |                                        |

Table 6.8 — Common CPO Function Overrides

# Chapter7 Graph Container Implementation

## 7.1 compressed\_graph

The `compressed_graph` is a high-performance graph container that uses **Compressed Sparse Row** format to store its vertices, edges and associated values. Once constructed, vertices and edges cannot be added or deleted but values on vertices and edges can be modified.

The following listing shows the prototype for the `compressed_graph`. Only the members shown for `compressed_graph` are public. No other member functions or types are exposed as part of the standard. All other types are only accessible through the types and functions in the Graph Container Interface. Multiple partitions (multi-partite) can be defined by passing the number of partitions in a constructor.

When a value type template argument (EV, VV, GV) is void then no extra overhead is incurred for it. The selection of the VId template argument impacts the inter storage requirements. If you have a small graph where the number of vertices is less than 256, and the number of edges is less than 256, then a `uint8_t` would be sufficient.

`compressed_graph` supports multipartite graphs.

All the `load` functions are supported, including `load_graph`, `load_vertices`, `load_partition` and `load_edges`.

```
template <class EV = void, // Edge Value type
          class VV = void, // Vertex Value type
          class GV = void, // Graph Value type
          integral VId = uint32_t, // vertex id type
          integral EIndex = uint32_t, // edge index type
          class Alloc = allocator<uint32_t>> // for internal containers
class compressed_graph {
public:
    compressed_graph();
    compressed_graph(size_t num_partitions); // multi-partite
    compressed_graph(const compressed_graph&);
    compressed_graph(compressed_graph&&);
    {tilde}compressed_graph();

    compressed_graph& operator=(const compressed_graph&);
    compressed_graph& operator=(compressed_graph&&);
}
```

[PHIL: Additional members to define?]

[PHIL: Add table capabilities/limitations]

[PHIL: tilde in destructor is giving problems in latex]



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